Suffa Academy

صُفّہ اکیڈمیؒ

STRIVE TOWARDS SUCCESS

CHEMISTRY 10 YRS UNSOLVED CHP WISE

CHEMISTRY XII

10 Year Questions Distribution According to the Chapters.

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QUESTIONS	YEARS
Chapter 1: INTRODUCTION OF FUNDAMENTAL CONCEPTS OF CHEMISTRY	
Section B	
iii) Write the group and periods of the following elements with Atomic numbers. * Z = 18 * Z = 24 * Z = 34 * Z = 47	2021
2. (i) (a) State the following laws: * Dobereiner's Law *Newlands law of Octaves 2. (i) (b) Identify the group, period and block of elements having atomic No.: *Z = 36 *Z = 24	2019
2. (i) Define Period and Group. Give the valence shell configurations of Alkali metals, Carbon family, Halogens and Noble gases.	2018
2. (iv) Write the electronic configuration and identify the period, group and block of elements bearing the following atomic numbers : 17,24,29,49.	2017
2. (i) Give the characteristics valence shell configuration of the following: *Zero group elements *Representative elements *Outer transition elements *Inner transition elements	2015
2. (i) What are the demerits of Mendeleev's periodic classification? How was it modified? 2. (vi)(a) Write the electronic configuration of Zn (Z=30) , group, period and block	2014
2. (i) (a) Give the valence shell electronic configuration of the following groups: *IIA and IB *IVA and VIB 2. (i) (b) Define the following with examples: *Dobereiner's Triads * Newland's Law of Octaves	2013
2. (i) State the Modern periodic law/. Explain how it removed the defects of Mendeleev's System of classification. 2. (ii) (a) Write the Valence shell configuration of the following groups. *IA and IB *VA and VB	2012
2. (i)(a) Write the electronic configuration, group, period and block of elements with atomic number 24 and 29. 2. (i)(b) Define:*Electron Population *Periodicity of elements	2011
Section C	
3. (c) Write the types of elements in the periodic table based on electronic configuration.	2019
3. (a) Define Moderan periodic law and discuss the different blocks in the modern periodic table.	2018
4. (c) State Mendeleev's periodic Law. Why this law was modified.	2017
3. (c) What are the types of elements based on electronic configuration in the periodic table?	2016
3. (a) Give the advantages and defects of Mendeleev's periodic table. Explain how modern periodic table is divide into blocks. Also write general electronic configuration of each block.	2011
2. (viii)(b) Write the electronic configuration of Zn (Atomic Number=30) & write its group, period and block	2010
Chapter 2: HYDROGEN	
Section B	
4.Define Binary compounds of hydrogen. Give their classification. Explain the preparation and properties of Ionic hydride.	2021
2. (ii) Write two methods for the separation of Hydrogen from Water Gas. Give the reactions of Atomic hydrogen with the following:*Phosphorus *Cupric Oxide	2019
2. (ii) What are isotopes? Explain the different isotopes of hydrogen.	2018
2. (vi)Give four Industrial methods for the preparation of Hydrogen gas (Except electrolysis of water.)	2017
2. (iv) (a) Define the term isotope. Explain the various isotopes of Hydrogen. 2. (iv) (b) Mention the simplest ions of Hydrogen and show their reactions with water. 2. (v) Describe two methods for the preparation of Water gas and give two methods for the separation of pure Hydrogen from it.	2016
2. (vi) What are Hydrides? How are they classified? Write the preparations and properties of Ionic hydrides.	2015
2. (vi)(b)Wat is Water gas? Give one method of its preparation.	2014
2. (ii) (a) Differentiate between the natural isotopes of Hydrogen.	2013

2. (ii) (b) Interstitial; Hydrides are not true chemical compounds. Comment.

2. (ii) Define and classify binary compounds of Hydrogen. Explain Ionic hydrides, giving two preparations and two reactions. — **2011**

2. (vii) (b) Give points of resemblance of Hydrogen with IA and VII-A groups of periodic table.

2. (i) (a) Explain that the position of hydrogen is misfit in Group IA of the periodic Table. — **2010**

2. (i) (b) What is water gas? Give one method of is preparation.

Section C

4. (b) What is meant by Binary Compounds of Hydrogen? Give their classification and discus the preparation and properties of salt-like hydrides. — **2019**

3. (c) Give the similarities and dissimilarities of Hydrogen with the elements of groups IA and VIIA. — **2018**

3. (b) What is meant by binary compound of Hydrogen, give their classification and give preparation and chemical properties of covalent hydrides. — **2017**

4. (b)Give the manufacture of Sulphuric acid by contact process. — **2016**

4. (a) Explain why Hydrogen is misfit in group IA and VIIA of the periodic table. — **2014**

2. (vi) Explain the extraction of Sodium metal by the electrolysis of molten chloride. — **2013**

2. (viii)How is Soda Ash manufactured by Ammonia Solvay process?

3. (a) What are binary compound of Hydrogen? Classify them. Explain covalent and Polymeric hydrides. — **2012**

3. (b) Writ down five industrial preparation of hydrogen gas. — **2010**

Chapter 3: s-BLOCK ELEMENTS

Section B

2 i) Refer to the list of given compounds. — **2021**

Compound	A	B	C	D
Specific Name	Carnallite	Blue Vitriol	Potash Alum	Barytes

* Write chemical formula of A and D.
* Write chemical formula of B and give chemical equation when it is heated at 230°C.
* Write three users of C.

v) Describe the extraction of sodium metal by Down's process.

2. (iii) Refer to the list of the given table — **2019**

Compound	A	B	C	D
Specific Name	Carnallite	Water Glass	Corundum	Lunar Castic

* Write the formulae of A and C.
* Write the equation for the preparation of B
* Wire the equation of reaction of D to heat at 450°C.
* Give any one use of D

2. (iii) Refer to the list of the given compounds: — **2018**

Compound	A	B	C	D
Specific Name	Epsom Salt	Plaster of Paris	Bleaching Powder	Baking soda

* Write the formulae of A and D.
* The equation for the preparation of B.
* Give only one common use of B.
* Give the equation for the reaction of C with HCl.

2. (vii) Why are the anode and cathode compartments separated in the extraction of sodium, describe the extraction of sodium by Down's process — **2017**

2. (viii) Refer the list of the given compounds:

Compound	A	B	C	D
Specific Name	Gypsum	Bleaching powder	Lunar Caustic	Potash Alum

Write :
* Formulae of "B" and "C"
* The equation for the preparation of B
* Given only one common use of "D"
* Given equation for the reaction of heating 'A' at 100°C

2. (iii) How is caustic soda prepared by Castner Kellner's Cell? Give advantages and disadvantages of the process — **2011**

Section C

4. (a) Describe the manufacture of Caustic soda by Castner-Kellner process (Diagram is not needed) — **2019**

3. (b) How is Soda ash manufactured by Ammonia Solvay process? Draw the flow chart also. — **2018**

[OR 3. (a)] How is caustic soda manufactured by the electrolysis of brine using moving Mercury as Cathode. Draw also the diagram of this method. — **2017**

4. (a) Give the industrial preparation of Sodium carbonate (Na_2CO_3) by Ammonia-solvay process. — **2016**

4. (b) Describe the extraction of Sodium from molten Sodium chloride.(Down Cell) — **2015**

3. (a) How is Soda ash manufactured by Ammonia Solvay process? — **2014**

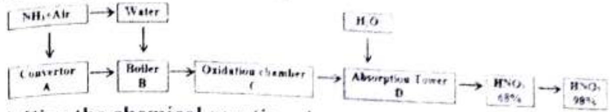
3. (a) How is sodium carbonate prepared by Ammonia-Solvay process? Draw the flow diagram.	2010

Chapter 4: p-BLOCK ELEMENTS

Section B

vi) What is Aqua regia? How does gold dissolved in it. Write its equation.	2021
2. (viii) Write about the manufacturing of Chlorine gas by Nelson Cell.	2019
2. (v) Give the refining of Aluminum by Hoop's electrolytic method (no diagram need.)	2018
2. (ii) What is Aqua Regia? How does gold dissolve in it? Give the reactions.	2016
2. (iv)(a) Give the geometrical structures of H_2SO_4 and H_2S. 2. (iv)(b) How is H_2S gas prepared in the laboratory? 2. (viii) Write the names and formulae of oxyacids of Boron and give the effects of change in temperature on them.	2015
2. (iii) Describe the manufacture of Chlorine gas by Castner-Kellner call. 2. (v) (b) How is Bauxite ore purified by Baeyer's or Serpeck's process? 2. (viii)(a) Draw the structure of HNO_3 in vapour and solid phases. 2. (viii)(b)What is Aqua Regia? How does Gold dissolve in it? Give the reaction.	2014
2. (iv) What is Allotropy? Describe amorphous form of Sulphur. Give reasons for its softness and elasticity.	2013
2. (ii) (b) What is Aqua Regia? How does gold dissolve in it? Give equations of the reaction. 2. (vi) Describe the preparation of Chlorine gas by Castner-Kellnear cell or Nelson cell.	2012
2. (ii) Write the names and formulae of any four ores of Aluminium. Give the refining of Aluminium by Hoop's Electrolytic method. 2. (v)(a) Give the composition & uses of two alloy of Aluminium. 2. (v)(b) Compare 4 properties each of diamond & graphite.	2010

Section C

3. What is metallurgy? Describe the extraction of 99.99% pure Aluminium from bauxite one containing SiO_2 and Fe_2O_3 as impurities.	2021
3. (a) Describe the extraction of 99.99 % pure Aluminium form Bauxite ore containing excess of Ferric oxide 4. [OR (a)] Describe Ostwald's Method for the manufacture of	2019
process with diagram.	2018
3. (a) The flow chart represents sages in the manufacturing of HNO_3 by Ostwald's Process:  * Give the chemical reactions in stages A,C,D with conditions. *Describe the conditions to get 95% oxidation of NH_3 and No and how 98% HNO_3 obtained. 4. (a) Write the different oxide ores of Aluminum. How Aluminum is obtained from bauxite ore containg Silica as impurity (Draw diagram of the cell)(Refining of Al is not required)	2017
3. (a) What is meant by Metallurgy. Describe the extraction of 99.99 % pure Aluminium form Bauxite ore containg SiO_2 and Fe_2O_3 as impurities. 4. (a) Describe the manufacture of Nitric acid by Ostwald's process. Draw the flow diagram. Write balanced equations for the reaction of dilute and very dilute Nitric acid with Zn. 4. (c) How is Chlorine manufactured by Nelson Cell? Draw its diagram. Give equation for the Auto-oxidation reduction reaction of Chlorine.	2015
4. (b) Describe the manufacture of Sulphuric acid by Contact process. Draw the flow diagram	2014
3. (a) What s Metallurgy? Explain the extraction of 99.99% pure Aluminium from Bauxite ore containing silica as impurity. 3. (c) Give the oxidizing reactions of H_2SO_4 with metals and non-metals 4. (a) Explain the manufacture of stable oxyacid of Nitrogen by Ostwald's method. Write balanced equation for the reaction of HNO_3 with Cu and C.	2013
3. (b) Describe the manufactures of Sulphuric acid and Contact process. Draw the flow diagram 3. (c) Define Alum and write chemical formulae of three alums.	2012
3. (b) Give the Structure and preparation of HNO_3, with flow diagram by Ostwald's process. Write balanced equations for the reaction of concentrated and dilute HNO_3 with Cu. 4. (a) Give with diagram the extraction of pure Aluminium from pure Aluminium oxide. What are Ammonal and Aluminium bronze? Mention their uses	2011
3. (c) Write the chemistry of boric acid. 4. (a) Describe the manufacture of Sulphuric acid by Contact	2010

process. Write the conditions for getting a good yield of SO_3. Draw the flow diagram. 4. (c) What is aqua regia? How does it dissolve gold? Explain with equations.	

Chapter 5: d-BLOCK ELEMENTS (TRANSITION ELEMENTS)

Section B

2. (vii) Describe the refining of Copper by electrolysis of blister copper.	2019
2. (vii) Discuss any two of the following properties of Transition metals. *Magnetic properties *Colour *Formation of complexes [OR 2. (vii)] What is Corrosion? Write its causes. Give different methods for the prevention of metals from Corrosion.	2018
2. (i) How is copper Mattel obtained from its concentrated Sulphide Ore.	2017
2. (viii) With the help of Crystal field theory, explain the colour of Transition metal complex ions.	2016
2. (vii) How is Blister copper obtained from matte?	2015
2. (vii)(a) How does Crystal Field Theory explain the colour of complex ions?	2014
2. (viii)(a) What is Blue vitriol? Give two equations for its preparation. 2. (viii)(b) What are outer and inner transition elements and how many series of each of them are present in the periodic table? Also write the names of the inner transition series.	2012
2. (iv) (a) Discuss any two of the following properties of transition metals: *Magnetic Properties *Formation of Complexes *Formation of Coloured Compounds 2. (iv) (b) What is Lunar caustic? Give the equation for its preparation.	2011
2. (vi) Define Ligands/ Give the classification of ligands based upon the number of coordination atoms with examples. 2. (viii)(a) Write a note on any One of the following: *photography *Silvering of a Mirror	2010

Section C

[OR 3. (b)] Describe the metallurgy of blister Copper from concentrated pyrite ore.	2018
3. (c) Describe any two of the following properties of 'd' block element: *Magnetic properties *Catalytic properties	2017

*Formation of Colour *Variable oxidation state	
3. (a) Describe the extraction of 99.99% Copper from roasted pyrite ore.	2016
3. (b) Describe the extraction of pure Copper from roasted pyrite ore.	2014
4. (c) What is Corrosion? Write its causes. Give different methods for the prevention of metals from corrosion.	2013
4. (a) How is pure Copper obtained from roasted pyrite ore?	2012
4. (b) How is copper obtained from Matte? Explain the methods of its refining by electrolysis. Draw the diagrams.	2010

Chapter 6: INTRODUCTION TO ORGANIC CHEMISTRY

Section B

viii) Write the classification of organic compound with examples.	2021
2. (x) Write the classification of Organic compounds with examples.	2019
2. (ix) Define Isomerism. Give its types with examples.	2018
2. (xv) Write the classification of organic compounds with examples.	2015
2. (ix) Describe the natural sources of organic compounds.	2014
2. (x) Define Isomerism. Give its types with examples.	2011
2. (ix) (a) Define the following with examples: *Polymerization *Homologous Series	2010

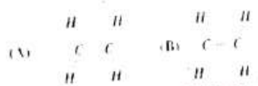
Section C

5. (b) Define Polymerization. How many types of Polymerizations are there? Give the preparation of following: *PVC *Bakelite	2015

Chapter 7: CHEMISTRY OF HYDROCARBONS

Section B

ix) The structure of two significance organic molecules A and B are shown below: (A) (B)  * Draw the hybrid orbital structure of 'B'	2021

* Give simple chemical test between A and B * Write equation for the preparation of 'A' from CH_3I. * Write chemical equation for reaction of 'B' with S_2Cl_2.	
2. (xi) Define Free radical. Give stepwise mechanism for the chlorination of Methane. 2. (xii) Draw the orbital structure of Ethylene and give equations for the formation of the following from Ethene: *Mustard gas *Glycol 2. (xiii) Why does Benzene give electrophilic substitution reaction? Give stepwise mechanism of nitration and chlorination in Benzene.	2019
2. (xi) Draw and explain the orbital structure of Acetylene. Give two methods of its preparation. 2. (xii) Give mechanism of these electrophilic substitution reaction of Benzene. *Friedel-Craft acylation *Sulphonation	2018
2. (x) The structure of two organic compounds 'A' and 'B' are shown below:  * Draw the hybrid orbital structure of 'A' showing bonds. * Write equation for the preparation of 'B' from CH_3I *Write equation for the reaction of A with S_2Cl_2.	2017
2. (x) Draw and explain the orbital structure of Acetylene. 2. (xii) Describe free radical mechanism reaction of Chlorination of Methane in presence of sunlight, giving all the equations involved.	2016
2. (x) Draw and explain the orbital structure of Ethene. 2. (xi) Write the Nitration reaction in Benzene along with its mechanism. 2. (xvi) Write the stepwise reactions for the following preparations. * m-Nitro benzoic acid from Benzene * Glyoxal from Benzene	2015
2. (ix) (a) Draw and explain the orbital structure of Ethyne. 2. (xi) (a) Give the mechanism of chlorination of Methane in the presence of sunlight. 2. (xv) (a) How is Benzene prepared commercially from petroleum?	2014
2. (x) (b) "Acetylene shows acidic properties" Give two reaction to justify this statement. 2. (xvi) (a) Give two reaction in which Benzene ring is not retained.	2013

2. (xi) (a) Give the reaction of Benzene with Chlorine: *In presence of Lewis acid catalyst * In presence of sunlight 2. (xiii) (a) Draw and explain the orbital structure of Ethyne. 2. (xvi) (a) How does Chlorine react with Methane in the present of sunlight?	2012
2. (ix) (a) Draw and explain the orbital structure of Ethene. 2. (xv) (b) State Markownikoff's rule. Write the equation for the reaction between Vinyl bromide and Hydrogen bromide.	2011
2. (ix) (b) Draw the orbital structure of an Ethyne molecule. Briefly explain the nature of hybridization in carbon and their bonds. 2. (xi) (a) Write the nitration reaction in benzene along with its mechanism. 2. (xi) (b) Explain the acidity of Ethyne with the help of two chemical reactions.	2010
Section C	
5. Explain the molecular orbital treatment of benzene; write the reaction of benzene with H_2 and O_2.	2021
5. (a) Explain Molecular Orbital treatment of Benzene.	2019
5. (a) What is orientation in Benzene? Explain orientation in mono-substituted Benzene. Give the names of ortho-para and meta directing group [OR 5. (a)]Give Kekule's structure of Benzene. Write the objections against it. How were these objections removed?	2018
5. (a) Explain the structure of Benzene by molecular orbital treatment and discuss the stability of Benzene molecule.	2017
5. (c) Describe Kekule structure of Benzene. Write the objection against it. How was this objection removed by Kekule?	2016
5. (a)Explain the molecular orbital treatment of Benzene. Also draw the resonance structures of Benzene.	2015
5. (a)Describe Kekule structure of Benzene. Write the objection against it. How was this objection removed by Kekule? 6. (b) What does Benzene give Electrophilic Substitution reactions? Explain the mechanism of Friedel Craft's Acylation.	2014
5. (a) Explain the molecular orbital treatment of Benzene. Also draw the resonance structures of Benzene. 5. (c) Draw the molecular structure of Ethene and discuss its orbital structure.	2013
5. (c) What is Orientation in Benzene?? Explain orientation in mono substituted benzene. Name three each of ortho para and meta directing groups. 6. (a) Explain electrophilic substitution in Benzene with	2012

reference to: (i)Friedel-Craft's acylation and (ii) Sulphonation reactions	
6. (a) Explain the structure of Benzene by molecular orbital treatment along with orbital structures. Benzene acts as a saturated and unsaturated compound. Justify this statement by equations	2011
5. (a) Give the Kekule's structure of benzene & write the objection against it. How was this objection removed by Kekule?	2010

Chapter 8: ALKYL HALIDES

Section B

2. (xv) What is Organometallic compound? How is it prepared? Starting CH_3MgI prepare the following: * Ethane *Ethanoic Acid *Ter-Butyl alcohol	2019
2. (xiii) What is Grignard Reagent? Give in reactions with the following * CH_3-NH_2 * HCHO * CO_2 * $CH_3-CO-CH_3$	2018
2. (xvi) Why do 1° Alkyl halides give SN^2 mechanism while 3° Alkyl halides give SN^1 mechanism?	2016
2. (xiii) What are Organometallic compounds? Prepare the following from any one organometallic compound. *Ethane *Ethyl Alcohol *Vinegar	2015
2. (xiii) Outline the step-wise reaction mechanism of the following: *SN^2 reaction between Bromo methane and NaOH *SN^1 reaction between 2-Chloro-2-methyl propane and NaCN.	2013
2. (xvi) (b) What are Alkyl halides? Classify them with one example of each.	2012
2. (xii) How is Grignard's reagent prepared? How can the following be prepared form Grignard's reagent? *Acetic acid *Ethyl Alcohol *Propane	2011
2. (xii)(a) What is meant by Elimination reaction? Explain the mechanism of unimolecular elimination reaction (E_1) 2. (xvi) Write the chemical equations to show that products will be formed when CH_3MgCl reacts with: (i)CO_2 (ii)acetone (iii)Methyl chloride (iv)water	2010

Section C

5. (OR)--Explain SN^1 and SN^2 reaction mechanism with example.	2021
6. (a) What are Nucleophilic substitution reaction? Explain the reaction mechanism of SN^2 and SN^1 with example	2019
5. (c) What is β-Elimination reaction? Write the mechanism of E_1 and E_2 reactions.	2018

6. (a) What are nucleophilic substituting g reaction. Outline the stepwise reaction mechanism for the following (i) SN^2 reaction between Bromo methane and NaOH (ii) SN^1 eaction between 2-chloro-2-methyl propane and NaCN.	2017
6. (a) What is Elimination reaction? Explain the mechanism of E_1 and E_2 reactions.	2016
6. (a) What is Nucleophile? Explain the reaction mechanism of SN_1 and SN_2 reactions with example.	2015
5. (b) What are Elimination Reaction? Write Bimolecular Elimination Reaction with its mechanism.	2014
6. (b) Outline the stepwise reaction mechanism of the following: * SN^1 reaction between NaCN and 2-Chloro-2-methyl propane	2012
5. (c) Prepare phenol by Down's process. 6. (c) Explain the mechanism of SN^1 reaction and SN^2 reaction with examples	2010

Chapter 9: CARBON COMPOUNDS WITH OXYGEN CONTAINING FUNCTIONAL GROUPS

Section B

2. (vii)(b) Draw the structure of the following: *Chelating agent *Chelate 2. (xv) What is fermentation? Explain the manufacture of Ethanol from starch.	2016
2. (xiii) (a) What are Monohydric alcohols? How are they classifies? 2. (xiii) (b) How is wood spirit manufacture from water gas	2014
2. (xvi) (b) Give two preparation of dimethyl ketone	2013
2. (xiii) (a) Define Functional group. Name the functional groups in the following compounds: *C_2H_5COCl *CH_3CONH_2 2. (xv) (a) What are Phenols? Classify them?	2011

Section C

6. Define Fermentation. How is ethanol manufacture by the fermentation of the following. * Starch * Molasses	2021
6. (a) What is Fermentation? How is Ethyl alcohol manufactured by fermentation of *Starch *Molasses	2018
6. (a) What is fermentation? How is Ethyl alcohol manufactured by the fermentation of: *Starch *Molasses	2014
5. (a) What is Fermentation? How is Ethyl alcohol obtained by fermentation of: *Starch *Molasses	2012

6. (b) Define Fermentation. How is ethyl-alcohol prepared by the fermentation of starch? Explain how ethyl-alcohol is made unfit for drinking.	2010

Chapter 10: CHEMISTRY OF LIFE

Section B

2. (xiii) What are enzymes? Explain the various factors which influence the rate of enzyme action. 2. (xv) Write equations and give the mechanism of Alkylation and Acylation in benzene.	2017
2. (xiv)(a) Explain the essential Fatty acids.	2016
2. (xii) Write the role of Amino acids in the human body.	2015
2. (ix) (b) What is Rancidification? Mention its causes 2. (xi) (b) How is the purity of oil an d fat determined?	2014
2. (xv) (a) Give the biological importance of Carbohydrates. 2. (xv) (b) Explain saponification of oils and fats with the help of chemical equation. Write the names of products formed.	2013
2. (xii) Define and classify Vitamins. Explain vitamin B-Complex and Vitamin C.	2012
2. (xii)(b) Explain saponification of oils and fats with the help of chemical equations. Write the names of the products formed. 2. (xiv) Define and classify Vitamins. Explain Vitamin B-Complex and Vitamin C.	2011

Section C

5. (c) Define Amino acids. How are they classified? Explain the role of Amino acids in the human body.	2019
[OR 6. (a)] What are carbohydrates? How are they classified and discuss the biological importance of carbohydrates.	2018
[OR 5. (c)] What is amino acid? Describe the importance of amino acid in human life.	2017
6. (a) What are Vitamins? How are they classified? Write their names and sources. Also name diseases caused by their deficiency.	2013
5. (b) What are vitamins? How are these classified? Write the names, Sources & diseases caused by their deficiency.	2010

Chapter 11: Chemical Industries in Pakistan

Section B

2. (xiv)What is Synthetic fiber? Explain Nylon and Polyester.	2014
2. (xv)What is Fertilizers? Name tow nitrogenous & one phosphates fertilizer along with their methods of preparation.	2010

Section C

5. (c) What is synthetic fiber? Explain Nylon and Polyester.	2017
6. (b) What is Fertilizer? Give its types. Explain phosphatic fertilizers.	2015

MISCELLANEOUS QUESTIONS (INORGANIC CHEMISTRY)

I.U.P.A.C NAMING (CHAPTER 5)

QUESTIONS	YEAR
ii) Write the IUPAC names of any four of the following: * $K_3[Fe(C_2O_4)_3]$ * $[Co(CN)_6]^{-3}$ * $[Cu(NH_3)_4]SO_4$ * $[Ni(en)_2Cl_2]^{+2}$ * $[Cr(H_2O)_6](NO_3)_3$	2021
2. (vi) Write the I.U.P.A.C. names of the following complexes: *$[Co(en)_3](NO_3)_3$ *$[Cu(NH_3)_4]^{2+}$ *$[AuCl_4]^{-1}$ *$Na_3[Fe(CN)_5NO]$	2019
2. (vi) Write I.U.P.C. names of the following complexes: * $[Pt(NH_3)_2Cl_4]$ * $NH_4[Cr(SCN)_4(NH_3)_2]$ *$[Cu(CN)_4]^{-2}$ *$[Co(H_2N-CH_2-CH_2-NH_2)_3](NO_3)_3$	2018
2. (iii) Write the I.U.P.A.C. names of the following: *$[Pt(NH_3)Cl_2]$ *$[Fe(CO)_4]$ *$Na_3[AlF_6]$ *$[CoBr_2(en)2]Br$	2017
2. (i) Write I.U.P.A.C. names of the following: *$[Co(NH_3)_3Cl]$ *$[Cr(en)_3]Cl_3$ *$Na_3[Cu(OH)_2(NO_3)_2]$ *$Na_2[Fe(CN)_5NO]$	2016
2. (v) (a) Write I.U.P.A.C. nems of the following: *$[Fe(NO_2)_6]^{-3}$ *$[Cr(en)_2Cl_2]Cl$	2014
2. (vii) (a) Write I.U.P.A.C. names of the following: *$NH_4[Cr(SCN)_4(NH_3)_2]$ *$Na_3[Fe(CN)_5NO]$	2013
2. (v) (b) Write I.U.P.A.C. names of the following: *$[Fe(NO_2)_6]^{-3}$ *$[Cr(en)_2Cl_2]Cl$	2012
2. (viii) Name the complexes by I.U.P.A.C. system: *$[Ni(CO)_4]$ *$NH_4[Cr(NCS)_4(NH_3)_2]$ *$[Cr(NH_3)_6]$*$[Fe(CN)_6]^{-3}$	2011
2. (iii) (b) Write down the I.U.P.A.C. names of the following: (i)$K3[Fe(CN)_6]$ (ii)$[Co(NH_2-CH_2-CH_2-NH_2)_3]Cl_3$	2010

SCIENTIFIC REASONS

QUESTIONS	CH.	YEAR
2. (iv) Give scientific reasons for any Four of the following: *Boric acid is weak monobasic acid	4	2019
* Alkali meals form cations easily.	3	
* Plastic Sulphur is elastic.	4	
* Graphite conducts electricity parallel to the plane.	4	
*Transition metals form non-stoichiometric compounds.	5	
2. (iv)Give scientific reasons for any four of the following: * Diamond is non-conductor of electricity.	4	2018
* H_2O and NH_3act as ligands but H_3O^+ & NH_4^+do not.	4	
* Blue crystals of Copper sulphate turns white on heating.	5	
2. (ii) Give scientific reasons for the following: * Alkali metals cannot be used in votaic cell	1	2017
* Alkaline earth metals are harder than Alkali metals.	1	
* Boric acid is soft and silky.	4	
* Anhydrous $CuSO_4$ is white while hydrated $CuSO_4$ is blue	5	
2. (iii) Give scientific reasons for the following: *Zinc hydroxide is soluble in excess of Sodium hydroxide solution.	5	2016
*Alkali and alkaline earth metals from only +1 and +2 ions respectively	1	
*Atomic size of Sulphur is bigger than that of Oxygen	4	
* Plastic Sulphur is elastic.	4	
2. (ii) Give scientific reasons for the following: * Hydrogen exhibits +1 and -1 oxidation states in its compounds.	2	2015
* Alkali metals are powerful reducing agents.	1	2015
* SO_3 is dissolved in H_2SO_4 but not in water.	4	
* Transition elements show variable oxidation states		
2. (ii)) Give scientific reasons for any four: * NH_3 and H_2O acts as ligands but NH_4^+ and H_3O^+ do not	4	2014
*Graphite conducts electricity, whereas diamond does not.	4	
* B_2O_3 is acidic while Al_2O_3 is amphoteric	4	

* Alkaline earth metal ions are more strongly hydrated than alkali metal ions.	1	
* Most of the transition elements and their compound are paramagnetic.	5	
2. (v) Give reasons: * The elements of a group in the periodic table have the same valence shell electronic configuration.	1	2013
* Alkali metals cannot be used in voltaic cell.	1	
* In d block elements, 4s orbital are filled prior to 3d orbitals but 4s elections are lost first in ionization.	1	
*Ligands are generally called Lewis bases.	5	
2. (iii) Give the reason of the following: * Ionic Hydrides are called true Hydrides.	2	2012
* Lithium and Beryllium markedly differ from other members of their respective groups.	1	
* Plaster of Paris is used in making plaster coats and Moulds.	3	
* Why is electronic configuration of Chromium (Cr) $4S^1,3d^5$ instead of $4s^2,3d^4$ while that of Copper(Cu) is $4s^1,3d^{10}$ instead of $4s^2,3d^9$?	5	
2. (v)Give reasons: * Alkaline earth metal ions are more strongly hydrated than alkali metal ions.	2	2011
* Atomic Hydrogen is more reactive than molecular hydrogen.	2	
* Most of the transition elements and their compounds are paramagnetic.	5	
* H_2O and NH_3 , act as Ligands but H_3O^+ & NH_4^+ do not	4	
2. (iv) Give reasons: (a) The ionization potential increases with the increase in atomic number in the period but decreases in Group from up to down.	1	2010
(b) Why is Heavy Water heavy?	2	
(c) Nascent hydrogen is more reactive than molecular hydrogen.	2	
(d) The viscosity and boiling point H_2SO_4 are high.	4	

EQUATIONS (FROM INORGANIC SECTION)

QUESTIONS	YEARS

3. (OR) Complete and balance the following equations: * $Al_4C_3 + H_2O \rightarrow$ * $K + O_2 \rightarrow K + O_2$ * $Cr_2O_3 + KOH + Br_2 \rightarrow$ * $CuSO_4 + KI \rightarrow$ * $Al + KOH + H_2O \rightarrow$ * $S + HNO_3 \rightarrow$ * $CH_4 + HNO_3 \rightarrow$ * $Fe_2O_3 + Al \rightarrow$	2021
3. (b) Complete and Balance the following equations: * $Cu + H_2SO_4 \longrightarrow$ * $K_2MnO_4 + Cl_2 \longrightarrow$ * $K_2Cr_2O_7 + KOH \longrightarrow$ * $HCOOH + H_2SO_4 \xrightarrow{Conc.}$ * $CuFeS_2 + O_2 \longrightarrow$	2019
4. (b) Give complete and balanced equations of any five of the following: * $K_2CrO_4 + H_2SO_4 \rightarrow$ * $CuSO_4 + KI \rightarrow$ * $C_6H_{12}O_6 + H_2SO_4 \xrightarrow{CONC}$ * $Mg + HNO_3 \xrightarrow{dilute}$ * $FEO.Cr_2O_3 + K_2CO_3 + O_2 \rightarrow$ * $Fe_2O_3 + Al \rightarrow$	2018
4. (b) Give complete and balanced equation(Any five) (i) $CaOCl_2+CO_2+H_2O \longrightarrow$ (ii) $Al+Fe_2O_3 \longrightarrow$ (iii) $K_2Cr_2O_7+KCl+H_2SO_4 \longrightarrow$ (iv) $P+HNO_3 \longrightarrow$ (v) $Sb_2S_3+HCl \longrightarrow$ (vi) $Zn+H_2SO_4 \longrightarrow$	2017
3. (b) Complete and balance the following equation: (i) $K_2Cr_2O_7 + H_2SO_4 \longrightarrow$ (ii) $KCl + K_2Cr_2O_7 + H_2SO_4 \longrightarrow$ (iii) $NaOH+Cl_2 \longrightarrow$ (iv) $P+HNO_3 \longrightarrow$ (v) $K_2MnO_4 + Cl_2 \longrightarrow$	2016
3. (c) Complete the following equations: (i) $Zn + H_2SO_4 \longrightarrow$ (ii) $AgBr + Na_2S_2O_3 \longrightarrow$ (iii) $P+HNO_3 \longrightarrow$ (iv) $Ca_2B_6O_{11}+Na_2CO_3 \longrightarrow$ (v) $K_2Cr_2O_7 + Fe\ SO_4 + H_2SO_4 \longrightarrow$	2015
3. (c) Complete the following equations: (i) $P+HNO_3 \longrightarrow$ (ii) $Zn + HNO_3 \longrightarrow$ (iii) $H_2S + H_2SO_4 \longrightarrow$ (iv) $CaOCl_2+CO_2+H_2O \longrightarrow$ (v) $AgBr + Na_2S_2O_3 \longrightarrow$	2014
3. (b) Complete the following Equation (i) $FeO.Cr_2O_3+K_2CO_3+O_2 \longrightarrow$ (ii) $KMnO_4+Cl_2 \longrightarrow$ (iii) $CuSO_4+KI \longrightarrow$ (iv) $Zn + HNO_3 \longrightarrow$ (v) $MgCO_3 + H_2SO_4 \longrightarrow$	2013
4. (b) Complete the following equations: * $Zn + HNO_3 \longrightarrow$ * $Ca_2B_6O_{11} + NO_3 \longrightarrow$ * $Cl_2+NaOH \longrightarrow$ * $CaSO_4.2H_2O \longrightarrow$	2012

2. (vii)Complete the following reaction (a) $Na_2B_4O_7+7H_2O \longrightarrow$ (b) $4Zn+10HNO_3 \xrightarrow{dil}$ (c) $HCOOH+H_2SO_4 \xrightarrow{conc}$ (d) $K_2Cr_2O_7+2H_2SO_4 \longrightarrow$	2010

COMPLETE AND BALANCED EQUATION

QUESTIONS	YEARS
iv) Write equation for the following reactions: * Reaction of SO_2 with Cl_2 * Heating MgO with carbon and Cl_2 gas. * Reaction of Mg with cold and dilute HNO_3 * Reaction of Aluminium with conc H_2SO_4	2021
. (v) Give complete and balanced Equations for any Four of the following: * Saturated solution of Soda ash treated with Carbon dioxide. * Action of superheated Water on Boron nitride. * Reduction of Sulphric acid with Hydrogen Sulphide. * Reaction of Sulphur dioxide with Chlorine gas. * Blue Vitriol heated to 230°C.	2019
2. (viii) Give complete and balanced equation for any four of the following: *Action of NaOH on Carbon monoxide. *Action of Concentrated HNO_3 on Sulphur. *Potassium manganite is treated with Chlorine * Colemanite is treated with Sodium Carbonate solution. * Action of Hydrogen sulphadie on Ferric chloride.	2018
2. (v) Give complete and balanced equations for the following reactions *Soda Ash heated with Silica *Boric acid is reacted with Sodium Carbonate *Litharge is heated with excess of air *Aluminum is reacted with Sodium hydroxide	2017
2. (vi) Write equations for the following reactions: * Conc. Suphuric acid with Oxalic acid *Nitric acid with Sulphur *Fe^{+3} wihH_2S * Blue stone treated with KI	2016
2. (v) Give chemical equations for the following: * Reaction of Aluminium with conc. H_2SO_4. * Reaction of Sodium carbonate with Silica * Reaction of Caustic soda with Ammonium Sulphate. * Reaction of bleaching powder with atmospheric carbon dioxide and moisture.	2015
2. (iv)Give equation for the following reactions: * Chrome yellow with Caustic soda. * Electrolytic oxidation of Potassium manganite by water *Potassium dichromate with Conc. H_2SO_4.	2014

* Chromium oxide with KOH and Bromine water.	2013
2. (iii) Give equations for the following: * Reaction of Ferric oxide with Aluminium. * Reaction of Litharge with Sodium chloride * Reaction of Benzene with Sulphuric acid * Reaction of Sodium dichromate with Potassium chloride	
2. (iv) Give equations for the following: * Zinc is treated with 90% Conc H_2SO_4. * Action of Heat on KMnO. * Aluminum reacts with aqueous Sodium Hydroxide * Lunar Caustic is heated 450°C	2012
2. (vi)Give equations for the following: * Reaction of Na_2CO_3 with Silica * Reaction between Sodium ann Oxygen * Action of heat on Blue vitriol * Reaction between K_2CrO_7, and KOH	2011

NOTE

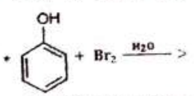
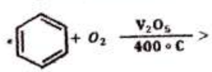
QUESTIONS	CH.	YEAR
4. (c) Write notes on any two of the following: *Photography *Silvering of mirror	5	2019
*Bleaching Powder	3	
*Borax	4	
4. (c) Write notes oon any two fo the following: *Aqua Regia	4	2018
*Tin plating *Silvering of Mirrors	5	
4. (c) Write notes on the following: *Tin plating	5	2016
*Borax	4	
3. (b) Write note on any One of the following: * Lead Pigments	4	2015
*Corrosion and its preventions	5	
*Lunar Caustic	5	
4. (c) Write notes on any two of the following: *Borax	4	2014
*Atomic Hydrogen	2	
*Blue vitriol	4	
4. (b) Write note on any Two of the following: *Paints *Plastic	11	2013
*Photography	5	

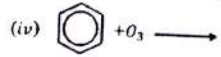
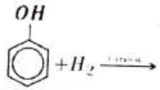
*Lead pigments	4	
2. (v) (a) Write a note on any one Detergents	11	2012
2. (v) (a) Write a note on any one OR Tin plating.	5	
4. (b) Write notes on any two of the following: *Lead pigments *Allotropic forms of Carbon	4	2011
*Corrosion and its prevention	5	

CHEMICAL FORMULAS

QUESTIONS	FORMULAE	YEAR
2. (vii)(a) Write the chemical formulae of the following: *Suhaga	$Na_2[B_4O_5(OH)_4]\cdot 8H_2O$	2018
*Alunite	$KAl_3(SO_4)_2(OH)_6$	
*Murda sang	PbO	
*Chromite ore	$FeCr_2O_4$	
2. (vii)(b)Write the formulae for the following: *Alunite	$KAl_3(SO_4)_2(OH)_6$	2014
*Hypo	$Na_2S_2O_2.5H_2O$	
*Magnesite	$MgCO_3$	
*Fluorspar	CaF_2	
2. (vii) (b)Write the chemical formulae of the following: *Tincal	$Na_2B_4O_7.10H_2O$	2013
*Lead sesquioxide	$O_3Pb_2^{-4}$	
*Stibnite	Sb_2S_3	
*Carnallite	$KMgCl_3\cdot 6(H_2O)$	
4. (c) Write the chemical formulae of the following: *Gypsum	$CaSO_4.2H_2O$	2012
*Suhaga	$Na_2[B_4O_5(OH)_4]\cdot 8H_2O$	
*Sandhur	Pb_3O_4	
*Lunar caustic	$AgNO_3$	
*Bleaching powder	$Ca(ClO)_2$	
2. (iii) (a) Write the chemical formulae of the following: *Plaster of Paris	$CaSO_4.1/2H_2O$	2010
*Baking Soda	$NaHCO_3$	
*Potash Alum	$KAl(SO_4)_2$	
*Oleum	$H_2SO_4.O_3S$ or $H_2O_7S_2$	

MISCELLANEOUS QUESTIONS
(ORGANIC CHEMISTRY)
COMPLETE THE EQUATIONS

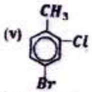
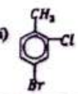
QUESTIONS	YEARS
6. (OR)--Complete the following reactions: * $C_2H_5OH + SOCl_2 \xrightarrow{\text{propane}}$ * $CH_3-CH_2OH \xrightarrow[180^\circ C]{Cu-Ni}$ * $CH_3-\overset{\overset{O}{\parallel}}{C}-CH_3 + H_2 \xrightarrow{Pd}$ * $CH_3-\overset{\overset{O}{\parallel}}{C}-CH_3 + 3I_2 + 2Na_2CO_3 \longrightarrow$ * $HCHO + Cu(OH)_2 + NaOH \longrightarrow$ * $HC \equiv CH + [O] \xrightarrow[Cold]{KMnO_4}$ * $C_2H_5OH + CH_3COOH \xrightarrow{H_2SO_4}$ * $CH_3-\underset{\underset{Cl}{\vert}}{CH}-CH_3 + Na \longrightarrow$ * $CH_3COOH \xrightarrow[500^\circ C]{MnO_2}$	**2021**
6. (b) Complete any five of the following reaction * $HC \equiv CH + HBr \longrightarrow$ * $CH_3-\overset{\overset{O}{\parallel}}{C}-CH_3 + 4[O] \xrightarrow[H_2SO_4]{K_2Cr_2O_7}$ *  * $CH_3COOC_2H_5 + NaOH \xrightarrow{10\%}$ *  * $HCHO + [Ag(NH_3)_2]OH \xrightarrow{Reflux}$	**2019**
6. (c)Complete and balance the following equations:	**2018**

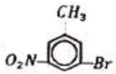
QUESTIONS	YEARS
(i) $CH \equiv CH \xrightarrow[Catalyst]{Organo-Nickel}$ (ii) $HCHO + NaOH \longrightarrow$ (iii) $CH_3COONa + NaOH\ CH \xrightarrow{CaO}$ (iv)  (v) $2CH_3-\overset{\overset{O}{\parallel}}{C}-OH \xrightarrow[500^\circ C]{MnO_2}$	
2. (iii) Complete the following reactions and give I.U.P.A.C. names of the products. * $AgOH + NH_4ON \rightarrow$ * $Zn(OH)_4 + NaOH \rightarrow$	**2015**
2. (xv) (b) Complete the following  $H-\overset{\overset{O}{\parallel}}{C}-H \xrightarrow{Polymerization}$	**2014**
2. (vii) Complete the following equation and give I.U.P.A.C. names of the complexes formed: * $CuSO_4 + NH_3 \xrightarrow{excess}$ * $AgNO_3 + NH_3 \xrightarrow{excess}$	**2012**
6. (a) Complete the following chemical process (i) $HC \equiv CH + 4[O] \xrightarrow{KMnO_4}$ (ii) $NaOH + CH_3COONa \xrightarrow{CaO}$ (iii) $2CH_3-CH_2-Cl + AgNO_3 \xrightarrow{Heat}$ (iv) $2HCHO + NaOH \longrightarrow$ (v) $(CH_3COO)_2Ca \xrightarrow{Dry\ distillation}$	**2010**

I.U.P.A.C NAMING

QUESTIONS	YEARS
x) Write I.U.P.A.C names any four of the following: * $(CH_3)_2-CH-(CH_2)_2-CH(CH_3)_2$ * $CH_3-(CHOH)_2-CH_2OH$ * $CH_2=CHCHClCH_2C=CH$ * $CH_2=CH-CH_2CHO$ * $CH_3-CH_2COCH(CH_3)_2$	**2021**
2. (xiv) Wire I.U.P.A.C names for any Four for the following:	**2019**

• $CH_3-\underset{}{\overset{OH}{\overset{\vert}{CH}}}-CH=CH_2$ • $CH_3-CH_2-\overset{OH}{\overset{\vert}{CH}}-CH_2-COOH$ • $CH_3-CH_2-\overset{O}{\overset{\Vert}{CH}}-CH_2-COOH$ • $C_2H_5-O-CH_2-C(CH_3)_3$ • 	
2. (xiv) Write I.U.P.A.C. names for any four for the following: (i) $CH_2=CH-C\equiv C-CH=CH_2$ (ii) $CH_2-(CH_2)_3-COOC(CH_3)_3$ (iii) $COH-(CH_2)_3-\underset{CH_3}{\underset{\vert}{CH}}-CH_3$ (iv) $CH_3-\underset{OH}{\underset{\vert}{CH}}-CH_2-COOH$ (iv) $CH_3-CO-CCl_3$	**2018**
2. (xii) Write I.U.P.A.C. Name of the following * $CH_2=CH-CH_2-COOH$ * $C_2H_5-(CH_2)_3-COCH(CH_3)_2$ * $CH_3-C(CH_3)_2-COCH(CH_3)_2$ * $CH_3-CH_2-OC(CH_3)_2C_2H_5$	**2017**
6. (c) Write the I.U.P.A.C. names of the following: (i) $(C_6H_5)_3CBr$ (ii) $C_2H_5OCH(CH_3)_2$ (iii) $H_2C=CH-\overset{Cl}{\overset{\vert}{CH}}-CH_2-\overset{CH_3}{\overset{\vert}{CH}}-C\equiv CH$ (iv) $(CH_3)_2CH-\overset{O}{\overset{\Vert}{C}}-C(CH_3)_3$ (v) $CH_3-\overset{O}{\overset{\Vert}{C}}-CH_2-COOH$	**2016**
6. (c) Write I.U.P.A.C. names of the following (i) $(CH_3)_3C-CH_2-CO-CH(CH_3)_2$ (ii) $CH_3-CH_2-\overset{O}{\overset{\Vert}{C}}-CH_2-COOH$ (iii) $CH_3-(CH_2)_3-\overset{O}{\overset{\Vert}{C}}-O-C(CH_3)_3$ (iv) $CH_3-O-\underset{OC_2H_5}{\underset{\vert}{CH}}-CH_2-\underset{OC_6H_5}{\underset{\vert}{CH_2}}$ (v) $CH_2=CH-C\equiv C-CH=CH_2$	**2015**
6. (c) Write I.U.P.A.C. names of the following:	**2014**

(I) $CH_2-\underset{CH_2}{\underset{\vert}{C}}-CH\equiv CH$ (II) $CH_3-\overset{O}{\overset{\Vert}{C}}-O-C(CH_3)_3$ (III) $CH_3-CH=CH-\underset{OH}{\underset{\vert}{CH}}-\underset{CH_3}{\underset{\vert}{CH}}-CH_3$ (IV) $CH_3-CH_2-O-C(CH_3)_3$ (V) 	
2. (xii) Name the following compounds by I.U.P.A.C. system: *$(CH_3)_2C.Br.CHO$ *$CH_3(CH_2)_3.COOC(CH_3)_3$ *$(CH_3)_3C.C.OH$ *$(CH_3)_2CHCOC(CH_3)_3$	**2013**
6. (c) Write the I.U.P.A.C. names of the following organic compounds: 6(c) (i) $CH_2=\underset{CH_3}{\underset{\vert}{C}}-C\equiv CH$ (ii) $CH_3-\overset{O}{\overset{\Vert}{C}}-O-\underset{CH_3}{\overset{CH_3}{C}}-CH_3$ (iii) $CH_3-CH=CH-CH_2-COOH$ (iv) $CH_3-CH=CH-\underset{OH}{\underset{\vert}{CH}}-\underset{CH}{\underset{\vert}{CH}}-CH_3$ (v) $CH_3-CO-CH_2-\underset{CH_3}{\underset{\vert}{CH}}-CH_3$ (vi) $CH_3-CH_2-O-\underset{CH_3}{\underset{\vert}{CH}}$ (vii) 	**2012**
6. (b) Write the I.U.P.A.C. names of the following organic compounds:	**2011**

(i) $CH_3-CH_2-CH=CH-COOH$ (ii) $CH_2=CH-C\equiv C-CH-CH_2$ (iii) $CH_3-\overset{\overset{O}{\parallel}}{C}-O-C_6H_5$ (iv) $CH_3-O-CH_2-\overset{\overset{OCH_3}{\mid}}{CH}-CH_2-OC_2H_5$ (v) $(CH_3)_2CH-\overset{\overset{O}{\parallel}}{C}-CH(C_2H_5)_2$ with CH_3 below (vi) $H-\overset{\overset{O}{\parallel}}{C}-(CH_2)_2-\overset{\overset{O}{\parallel}}{C}-H$ (vii) 	
2. (xiii)Write the I.U.P.A.C. names of the following: (a) $CH_3-CH=CH-CH_2OH$ (b) $HOOC-CH_2-CH_2-COOH$ (c) $HC\equiv C-CH_2-CH_2-CH=CH_2$ (d) $CH_3-\underset{\underset{O}{\parallel}}{C}-O-CH(CH_3)_2$	2010

DEFINITIONS

Questions	CH	Year
vii) Define the following: * Polymerization * Homologous series	6	2021
* Catenation	--	
* Glycosidic Linkage	10	
2. (ix) Define the following: *Homologous Series *Metamerism	6	2019
*Peptide bond *Rancidification	10	
2. (x)Define the following: *Saponification	10	2018
*Cracking *Polymerization	6	
*Catenation	--	
2. (ix) Define the following: *Functional group	6/9	2017
*Catenation	--	
*Glycosidic linkage *Saponification	10	
2. (ix) Define the following terms: *Cracking *Isomerism	6	2016
*Saponification	10	
2. (ix) Define the following with examples: * Octane number	1	2015
*Catenation	--	
*Functional group *Metamerism	6	
2. (x) Define the following: *Homologous series *Cracking *Refining of petroleum	6	2014
*Carbohydrate	9	
2. (x) (a) Define the following, giving examples: *Isomers *Polymers	6	2013
2. (ix) Define: *Zwitter Iron *Catenation	--	2012
*Polymerization *Metamerism	6	

PREPARATION/REACTION

QUESTIONS	YEARS
Section C	
2. (xvi) Wrie equations for the preparation of any four of the following compounds: * Oxime from Formaldehyde * Phenol from Chlorobenzene * Acetic anhydride from Sodium ethanoate * Red solid from Acetylene * Diethyl ether from Chloroethane	2019
2. (xv) Give the preparation of the following: * Carbolic acid from Benzene * Acetal from Methanal *Ethanal from Acetylene * Diethyl ether from Ethyal alcohol	2018
2. (xi) Write equations for the following reactions: *Cannizaro reaction *William Son Synthesis *Fermentation of Molasses *Esterification	2017
2. (xiv)How does Ethyl iodide react with the following reagents? * Sodium Ethoxide *Mg *KOH(Alcoholic) *Na	
2. (xiv) Write equation for the preparation of the following: * Phenol from Chlorobenzene * Ethyl ethanoate from Ethanol * Water gas from methane * Ethene from Ethyl bromide	2015
2. (xii) Give equations for the following reactions: * methanol with steam * Glucose with Tollen's Reagent * Acetyl chloride with Sodium ethanoate * Ethanol with Grignard Reagent	2014
2. (xiv) How will you prepare the following? * Carbolic acid from Benzene * Acetaldehyde from Acetylene * Acetal from Methanal	2013

*An Oxime from Formaldehyde	
2. (xi) (b) Write equation for the reactions of Alcoholic KOH with: *2-Bromo propane *2-Bromo-2-methyl propane	2013
2. (xi) (a) Which alkane is obtained by the reaction of metallic Sodium with the following Compounds? *2-Bromo-propane *2-Bromo-2methyl propane	2013
2. (x) Write equation only for the for the following conversions: *Benzene to Nitro-Benzene *Ethyle to Ethanal *Ethanol to Ethanoic acid	2012
2. (xv) Give equations for the following reactions: * Reaction of Sodium benzoate with Soda-lime * Reaction of Acetone with acidified $K_2Cr_2O_7$. * Reaction of Sodamide with Chloroethane * Reaction of Phenol with Conc. HNO_3.	2012
2. (vii) (a) Give the equations for the preparation of the following; *Bleaching Powder *Boric acid	2011
2. (xi) Give the equations for the preparation of the following: * Acetaldehyde form Acetylene * Ethene from Ethyl alcohol * Oxalic acid from Ethyne *Benzene from Toluene	2011
2.(x) Give the equations for the preparation of the following: *Ethane from methyliodie *Benzene from phenol *Ethene from ethylacohol *Ethyne from calcium carbide	2010

Section C

6. (b) How the following conversing carried out (i) Acetone into ter.butyl alcohol(ii) Ethanol into chloro ethane (iii) Diethyl ether inot Ethyl iodide (iv) Formaldehyde into meta formaldehyde (v)Acetyl chloride to Acetic anhydride. — **2017**

6. (c) Complete the following reactions:

(i) $CH_2 = Ch - CH_3 + HBr \longrightarrow$

(ii) $H - C \equiv C - H + H_2O \xrightarrow{HgSO_4 / H_2SO_4}$

(iii)  $+ O_2 \xrightarrow[400^\circ C]{V_2O_5}$

(iv)  $+ HNO_3 \xrightarrow[Heat]{Conc.}$

5. (a) Give the equations of the following: — **2016**
* Reaction of Ethanal with Grignard reagent
* Decomposition of Acetic acid in the presence of MnO_2
* Reaction of Methanal with caustic soda.
* Reaction of Chloromethane with Sodium ethoxide.

5. (b) Give the preparation of the following: * 2-Bromo propane from 1-Bromo propane * Phenol from Benzene * Benzoic acid from Benzene *Isopropyl alcohol from 1-Propanol	
5. (c) Give equation for the following reactions: * Phenol with dilute HNO_3 * Ethanal with Sodium and Ethanol * Toluene with $KMnO_4$. * 2-Propanone with Methyl magnesium chloride. * Acetylene with cold $KMnO_4$.	2014
5. (b) Prepare the following compounds from CH_3Br: * Methanol *Methyl thanoate *Methoxy ethane *Amino methane *Methl mercaptan (Methyl thioalcohol) 6. (b) Prepare the following compounds: *m-nitro benzoic acid from Benzene * Ethyl alcohol from Molasses	2013
5. (a) Give equations for the preparation of following organic compounds: (i)Ethyne from Ethene (ii)Diethyl ether from Ethanol (iii)Methanol from Water gas (iv)Methanal from Methanol (v)Acetone from Acetic acid (vi)Ethanoic acid from Ethanol (vii) Ethanol from Glucose	2011
5. (b) Give equation for the following reaction: (i)Cannizzaro's Reaction (ii)Polymerization of Acetaldehyde (iii) Reduction of Acetone (iv)Esterification (v)Combustion of Ethane (vi)Oxidation of Benzene(vii) Hydrolysis of Ethyl acetate in basic medium	2010

NOTE

Questions	CH	YEARS
6. (c) Write note on any one of the following: *Detergents *Paints * Glass	11	2019
6. (b) Write note on any one of the following: *Detergents *Fertilizers *Plastics	11	2018
6. (b) Write note on any one of the following: *Detergents *Plastics	11	2016
[OR 6. (b) | Write notes on any two of the following: *Glass *Detergent	11	2014
*Amino acids	10	2014
2. (xvi) Write notes on any two of the following: *Glass *Detergents *Plastic	11	2011

DIFFERENCES

QUESTIONS	CH	YEARS
2. (xvi) Differentiate between the following: * Saturated and Unsaturated Hydrocarbon	6	2014
* Reducing and Non-reducing sugar.	10	2014
2. (xiv) Differentiate between the following: *Saturated and Unsaturated Hydrocarbon * Aliphatic Hydrocarbon and Aromatic Hydrocarbon	6	2012

CHEMICAL TEST

QUESTIONS	YEARS
2. (xvi) Give a simple chemical test to distinguish between the following: *Alkane and Alkene *Ethene and Ethyne *Alkyl halide and Alkane *Aldehyde and ketone	2018
2. (xvi) Distinguish by simple chemical test only two of the following: *Ethyne and Ethene *Ethane and Ethene *Alkyl Halide and Alkane *Formaldehyde and Acetone.	2017
2. (xi) Give chemical test to distinguish between the following: *Aldehyde and Ketone *paraffin and Olefin *n-Hexane and Benzene *1-Butyne and 2-Butyne	2016
6. (c) Distinguish by simple chemical test *Hexane and Benzene *Aldehyde and Ketone	2013
2. (xiii) (b)Give chemical test to distinguish between the following: *Ethene and Ethyne *Aldehyde & Ketone	2012
2. (ix) (b) Distinguish by simple chemical tests: *Aldehyde and Ketone *Alkene and Alkyne	2011

WHAT HAPPENS

QUESTIONS	YEARS
Section B	
2. (viii) What happens when? * Ethylene react with cold aqueous $KMnO_4$ solution. * Mehtanol reacts with steam *2-Propanone is oxidized in presence of $K_2Cr_2O_7$ and conc. H_2SO_4. * Ethyl chloride reacts with Sodium metal.	2016
2. (xiv) What happens when the following reaction take place? * Methanl is reacted with Tollen's Reagent. * Acetone is heated with iodine and Na_2CO_3. * Hydrolysis of ethylacetate in basic medium *Acetic acid is heated with C_2H_5OH in presence of conc. H_2SO_4.	2010
Section C	
5. (c) What happens when: * Acetone is treated with I_2 in in the presence of Sodium carbonate. *Methyl iodide is treated with Sodium metal. * Excess Ethanol is treated with Conc. H_2SO_4. *Phenol is treated with Bromine water.	2015
5. (b) What happens when: * Methanal is treated with Phenyl hydrazine. * Phenol is treated with Bromine water. * Phenol is treated with red hot Zinc dust. * Ethyne is reacted with Iodine in presence of Ethanol	2012

STRUCTURAL FORMULAE

QUESTIONS	YEARS
Section B	
2. (xiv)(b) Draw the structure of the following *Lysine *Nicotinamide	2016
2. (xii)(b) Write the structural formula of the following: *Divinyl acetylene *Hydroquinone *Ethyl ter-butyl ether *2-Bromobutanal	2012
2. (xiii)(b) Give electronic structure of: *Acetone *Silver acetylide	2011
Section C	
5. (b) Draw the molecular structure of any Five of the following: *Catechol *Adipic acid *Nicotinamide *Picric acid *Isobutyraldehyde *Terephthalic acid	2019
5. (b) Draw the structural formulae of any five of the following: *p-Cresol *Picric acid *Valeric acid *2-bromo-3methyl butanal *Neo-pentyal alcohol *Di-siopropyl ether	2018
5. (b) Give the structure formula of the following compound. (Any five) (i)Pyrogallol (ii)P-Cresol (iii)Caproic acid (iv)β-methyl Butyraldehyde (v)Phenyl hydrazine (vi)Diethyal Acetylene	2017
6. (d) Give the structural formulae of the following: *Catechol *Divinyl acetylene *Glyoxal *Picric acid	2013

Solved Numericals of Chemistry

BY

M. Zubair Arif Yousufi

CHEMISTRY 2021

SECTION 'B'

INORGANIC CHEMISTRY

ii) Write the I.U.P.A.C. names of any four of the following:

* $K_3[Fe(C_2O_4)_3]$ * $[Co(CN)_6]^{-3}$
* $[Cu(NH_3)_4]SO_4$ * $[Ni(en)_2Cl_2]^{+2}$
* $[Cr(H_2O)_6](NO_3)_3$

ANSWER:

COMPOUND	I.U.P.A.C. NAMES
* $K_3[Fe(C_2O_4)_3]$	Potassium trioxalato Ferrate(III)
* $[Co(CN)_6]^{-3}$	Hexacyanidocobaltate(III)
* $[Cu(NH_3)_4]SO_4$	Tetrammine copper (II) sulphate
* $[Ni(en)_2Cl_2]^{+2}$	Dichloride bis-Ethylenediamine Nickel (II)
* $[Cr(H_2O)_6](NO_3)_3$	hexaaquachromium(III) nitrate trihydrate.

iii) Write the group and periods of the following elements with Atomic numbers.

* Z = 18 * Z = 24 * Z = 34 * Z = 47

ANSWER:

Atomic No.	Electronic Configuration	Group	Period	Block
18	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6$	VIII-A	"3"	"p"
24	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^5$	VI-B	"4"	"d"
34	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^4$	VI-A	"4"	"p"
47	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^1, 4d^{10}$	I-B	"5"	"d"

iv) Write equation for the following reactions:

* **Reaction of SO_2 with Cl_2**

$$SO_2 + Cl_2 \longrightarrow SO_2Cl_2$$

* **Heating MgO with carbon and Cl_2 gas.**

$$2MgO + C + 2Cl_2 \longrightarrow 2MgCl_2 + CO_2$$

* **Reaction of Mg with cold and dilute HNO_3**

$$4Mg + 10HNO_3 \xrightarrow{dilute} 4Mg(NO_3) + N_2O + 5H_2O$$

* **Reaction of Aluminium with conc H_2SO_4**

$$2Al + 6H_2SO_4 \xrightarrow{Conc.} Al_2(SO_4)_3 + 6H_2O + 3SO_2$$

ORGANIC CHEMISTRY

x) Write I.U.P.A.C names any four of the following:

Answers:

STRUCTURE	IUPAC NAME
* $(CH_3)_2 - CH - (CH_2)_2 - CH(CH_3)_2$	2,3 dimethyl Hexane
* $CH_3 - (CHOH)_2 - CH_2OH$	1,2,3 butane-triol
* $CH_2 = CHCHClCH_2C \equiv CH$	3 chloro-Hexaene-5-yne
* $CH_2 = CH - CH_2CHO$	3 hexene-1-al
* $CH_3 - CH_2COCH(CH_3)_2$	3 methyl pantane-3-one

SECTION 'C'

INORGANIC CHEMISTRY

3.(OR) Complete and balance the following equations:

* $Al_4C_3 + H_2O \rightarrow$ * $K + O_2 \rightarrow$
* $Cr_2O_3 + KOH + Br_2 \rightarrow$ * $CuSO_4 + KI \rightarrow$
* $Al + KOH + H_2O \rightarrow$ * $S + HNO_3 \rightarrow$
* $CH_4 + HNO_3 \rightarrow$ * $Fe_2O_3 + Al \rightarrow$

ANSWER:

* $Al_4C_3 + 12H_{2O} \rightarrow 4Al(OH)_3 + 3CH_4$
* $K + O_2 \rightarrow 2K_2O$
* $Cr_2O_3 + 10\,KOH + 3Br_2 \rightarrow 2K_2CrO_4 + 6KBr + 5H_2O$
* $2CuSO_4 + 4KI \rightarrow 2CuI + I_2 + 2K_2SO_4$
* $2Al + 2KOH + 6H_2O \rightarrow 2K[Al(OH)_4] + 3H_2$
* $S + HNO_3 \rightarrow H_2SO_4 + NO_2 + H_2O$
* $CH_4 + HNO_3 \rightarrow CH_3NO_2 + H_2O$
* $Fe_2O_3 + Al \rightarrow Al_2O_3 + Fe$

ORGANIC CHEMISTRY

6. (OR) Complete the following reactions:

ANSWER

* $C_2H_5OH + SOCl_2 \xrightarrow{propane} C_2H_5Cl + SO_2 + HCl$
* $CH_3 - CH_2OH \xrightarrow[180^\circ C]{Cu-Ni} CH_3 - CHO + H_{2(g)}$

* $$CH_3-\overset{\overset{O}{\|}}{C}-CH_3 + H_2 \xrightarrow{Ni} CH_3-\overset{\overset{O}{\|}}{C}-H-CH_3$$

* $$CH_3-\overset{\overset{O}{\|}}{C}-CH_3 + 3I_2 + 2Na_2CO_3 \longrightarrow CHI_3 + CH_3-\overset{\overset{O}{\|}}{C}-ONa + 3NaI + 2CO_2 + H_2O$$

$$HCHO + 2Cu(OH)_2 + NaOH \xrightarrow{warm} HCOOH + Cu_2O + 2H_2O$$

* $$HCHO + Cu(OH)_2 + NaOH \longrightarrow$$

OH OH

* $$HC \equiv CH + 4[O] \xrightarrow[Hot]{KMnO_4} O=C-C=O$$

* $$C_2H_5OH + CH_3COOH \xrightarrow{H_2SO_4} CH_3COOC_2H_5 + H_2O$$

* $$\underset{\underset{Cl}{|}}{CH_3-CH-CH_3} + Na \xrightarrow{Dry\ Ether} CH_3-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{CH_3}{|}}{CH}-CH_3$$

* $$2CH_3COOH \xrightarrow[500^\circ C]{MnO_2} CH_3COCH_3 + CO_2 + H_2O$$

CHEMISTRY 2019

SECTION 'B'

INORGANIC CHEMISTRY

2. (i) (b) Identify the group, period and block of elements having atomic No.:
*Z = 36 *Z = 24

Answer:

Atomic No.	electronic configuration	Group	Period	Block
24	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^5$	VI-B	"4"	"d"
49	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6$	VIII-A	"4"	"p"

2. (iii) Refer to the list of the given table

Compound	A	B	C	D
Specific Name	Carnallite	Water Glass	Corundum	Lunar Caustic

* Write the formulae of A and C.
Carnallite: $KMgCl_3 \cdot 6(H_2O)$
Corundum: Al_2O_3

* Write the equation for the preparation of B

$$Na_2CO_3 + SiO_2 \longrightarrow Na_2SiO_3 + CO_{2(g)}$$

* Wire the equation of reaction of D to heat at 450°C.

(iv) $2AgNO_3 \xrightarrow{450^\circ C} Ag + 2NO_2 + O_2$

* Give any one use of D

Large quantities of silver nitrate are used in the production of light sensitive plates, film and papers

2. (iv) Give scientific reasons for any Four of the following:

*Boric acid is weak monobasic acid

Answer: These acids have only one hydrogen atom to donate to a base in an acid base reaction just because they have one replaceable hydrogen atom. Due to this reason, they are known as monobasic.

* Alkali metals form cations easily.

Answer: Alkali metals lose valence electrons easily. the alkali metals have 1 valence electron. They need 8 Valence electrons to become stable, so it is easier to lose the 1 valence electron instead of gaining 7. When an element loses electrons, it becomes more positive, meaning it is a cation.

* Plastic Sulphur is elastic.

Answer: Plastic sulphur is an amorphous form of sulphur when liquid (molten) sulphur is poured in ice water the immediate freezing of material is responsible for irregular arrangement of atoms but a coils like structure is formed. The coiling and uncoiling of the structure is responsible for stretching or elastic nature of sulphur.

* Graphite conducts electricity parallel to the plane.

Answer: The flat plates of graphite are "connected" by intermolecular forces where there is no orbital overlap and no delocalized electrons.. Due to this reason, electrons cannot move from layer to layer, althought they are free to move within layers

*Transition metals form non-stoichiometric compounds.

Answer: Transition metals form non-stoichiometric compounds because Small atoms such as H, B, C and N can reside within the holes present in the crystal lattice of these elements and bear no stoichiometric composition.

2. (v) Give complete and balanced Equations for any Four of the following:

* Saturated solution of Soda ash treated with Carbon dioxide.

$$Na_2CO_3 + H_2O + CO_2 \longrightarrow 2NaHCO_3$$

* Action of superheated Water on Boron nitride.

$$Bn + 3H_2O \longrightarrow H_3BO_3 + NH_3$$

* Reduction of Sulphric acid with Hydrogen Sulphide.

$$H_2SO_4 + H_2S \longrightarrow SO_2 + H_2O + S$$

* Reaction of Sulphur dioxide with Chlorine gas.

$$SO_2 + Cl_2 \longrightarrow SO_2Cl_2$$

* Blue Vitriol heated to 230°C.

$$CuSO_4.5H_2O \xrightarrow{230^\circ C} CuSO_4 + 5H_2O$$

2. (vi) Write the I.U.P.A.C. names of the following complexes:

*$[Co(en)_3](NO_3)_3$ *$[Cu(NH_3)_4]^{2+}$ *$[AuCl_4]^{-1}$

*$Na_3[Fe(CN)_5NO]$

ANSWER:

*$[Co(en)_3](NO_3)_3$ **Tris (ethylene diamaine) Cobalt (III) nitrate**

*$[Cu(NH_3)_4]^{2+}$ **Tetra ammine copper (II) ions**

* $[AuCl_4]^{-1}$ **Tetra chloro aurte (III) ion**

*$Na_3[Fe(CN)_5NO]$ **Sodium penta cyano nitrosyl ferrate (II)**

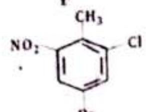
ORGANIC CHEMISTRY

2. (xiv) Wire I.I.P.A.C names for any Four for the following:

· $CH_3 - \underset{OH}{CH} - CH = CH_2$
3-Butene, 2-ol

· $CH_3 - CH_2 - \underset{OH}{CH} - CH_2 - COOH$
3-hydroxy pentanoic acid

· $CH_3 - CH_2 - \overset{O}{\overset{\|}{CH}} - CH_2 - COOH$
Iso propyl butanoate



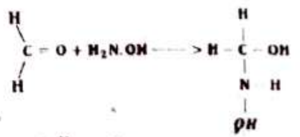
· $C_2H_5 - O - CH_2 - C(CH_3)_3$
2-ethoxy, 2-methyl propane

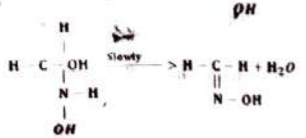
2 chloro, 4 bromo, 5 nitro toluene

2. (xvi) Write equations for the preparation of any four of the following compounds:

*** Oxime from Formaldehyde**

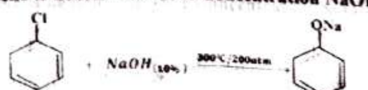
Formaldehyde adds hydroxalamine (NH_2OH) to form a stable product called "Oxime".

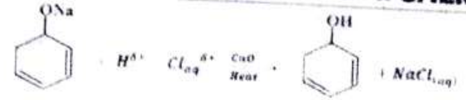




*** Phenol from Chlorobenzene**

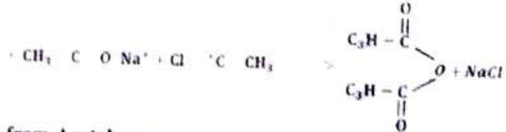
When Chlorobenzene is heated with 10% Concentration NaOH it gives Phenol





*** Acetic anhydride from Sodium ethanoate**

Acetic anhydride is obtained by reacting acetyl chloride with sodium acetate



*** Red solid from Acetylene**

When acetylene is heated with cuprous chloride in Ammonia cuprous acetylide (Red Solid) is formed

$$HC \equiv CH + Cu_2Cl_2 \longrightarrow CuC \equiv CCu + 2HCl$$

*** Diethyl ether from Chloroethane**

Diethyl ether is obtained by treating chloroethane with sodium ethoxide

$$CH_3CH_2Cl + C_2H_5ONa \longrightarrow C_2H_5 - O - C_2H_5 + NaCl$$

SECTION 'C'

INORGANIC CHEMISTRY

3. (b) Complete and Balance the following equations:

· $Cu + 2H_2SO_4 \longrightarrow CuSO_4 + SO_2 + 2H_2O$

· $2K_2MnO_4 + Cl_2 \longrightarrow 2KMnO_4 + 2KCl$

· $K_2Cr_2O_7 + 2KOH \longrightarrow 2K_2CrO_4 + H_2O$

· $HCOOH + H_2SO_4 \xrightarrow{Conc} CO + H_2O$

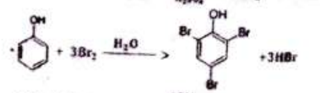
· $2CuFeS_2 + 4O_2 \longrightarrow Cu_2S + 2FeO + 3SO_2$

ORGANIC CHEMISTRY

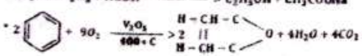
6. (b) Complete any five of the following reaction

· $HC \equiv CH + HBr \longrightarrow H_2C = \underset{|}{\overset{Br}{CH}}$

· $CH_3 - \overset{O}{\overset{\|}{C}} - CH_3 + 4[O] \xrightarrow[H_2SO_4]{K_2Cr_2O_7} CH_3 - \overset{O}{\overset{\|}{C}} - OH + CO_2 + H_2O$



· $CH_3COOC_2H_5 + NaOH \xrightarrow{10\%} C_2H_5OH + CH_3COONa$



· $HCHO + 2[Ag(NH_3)_2]OH \xrightarrow{Reflux} H - \overset{O}{\overset{\|}{C}} - ONH_4 + 2Ag + H_2O + 3NH_3$

CHEMISTRY 2018

SECTION 'B'

INORGANIC CHEMISTRY

2. (iii) Refer to the list of the given compounds:

Compound	A	B	C	D
Specific Name	Epsom Salt	Plaster of Paris	Bleaching Powder	Baking soda

* Write the formulae of A and D.

Formula of A: $MgSO_4.7H_2O$ Formula of D: $NaHCO_3$

* The equation for the preparation of B.

$$2CaSO_4.2H_2O(s) \xrightarrow{100^\circ C} > (CaSO_4)3.H_2O + 3H_2O$$

* Give only one common use of B.

COMMON USE OF PLASTER OF PARIS

Answer: It is used in surgery and castings.

* Give the equation for the reaction of C with HCl.

$$CaOCl_2 + 2HCl \longrightarrow > CaCl_2 + H_2O + Cl_2$$

2. (iv)Give scientific reasons for any four of the following:

* Diamond is non-conductor of electricity.

Answer: Diamond is non-conductor of electricity Because electricity is conducted in a crystal by electrons that are relatively free. But in diamond, each carbon atom is covalently bonded with four other carbon atoms and there are no free electrons.

* H_2O and NH_3act as ligands but H_3O^+ & NH_4^+ do not.

Answer: In H_2O and NH_3, there is a lone pair of electron which is available for donation. Therefore they act as ligands while in a case of H_2O^+ and NH_4^+ Ion pair of electron is not available-in fact it is consumed in making of coordinate covalent bond.

* Blue crystals of Copper sulphate turns white on heating.

Answer: Blue crystals of Copper sulphate turns white on heating, because water molecules are evaporate due to heat and it convert in hydrate to anhydrous $CuSO_4$. anhydrous $CuSO_4$ does not have water molecule in its compound it does not form d-d transition and due to which it will be no colour generally it is said to be as whit and whereas hydrated $CuSO_4$ will be in blue colour.

2. (vi) Write I.U.P.C. names of the following complexes:

* $[Pt(NH_3)_2Cl_4]$ * $NH_4[Cr(SCN)_4(NH_3)_2]$ * $[Cu(CN)_4]^{-2}$
* $[Co(H_2N-CH_2-CH_2-NH_2)_3](NO_3)_3$

ANSWER:

* $[Pt(NH_3)_2Cl_4]$: Diammine tetra chloro Platinum (IV)
* $NH_4[Cr(SCN)_4(NH_3)_2]$

Ammonium Diammine tetra thiocyanato Chromate (III)

* $[Cu(CN)_4]^{-2}$: Tetra Cyano Cuperate (II) ion
* $[Co(H_2N-CH_2-CH_2-NH_2)_3](NO_3)_3$

Tris (ethylenediamine) Cobalt (III) nitrate

2. (viii) Give complete and balanced equation for any four of the following:

*Action of NaOH on Carbon monoxide.

*Action of Concentrated HNO_3 on Sulphur.

*Potassium manganite is treated with Chlorine

* Colemanite is treated with Sodium Carbonate solution.

* Action of Hydrogen sulphadie on Ferric chloride.

ANSWER:

(i) $NaOH + CO \rightarrow HCOONa$

(ii) $6HNO_3 + S \rightarrow H_2SO_4 + +6NO_2 + 2H_2O$

(iii) $2K_2MnO_4 + Cl_2 \rightarrow 2KMnO_4 + 2KCl$

(iv) $Ca_2B_6O_{11} + 2Na_2CO_3 \rightarrow 2CaCO_3 + Na_2B_4O_7 + 2NaBO_2$

(v) $H_2S + 2FeCl_3 \rightarrow 2FeCl_2 + S + 2HCl$

ORGANIC CHEMISTRY

2. (xiv) Write I.U.P.A.C. names for any four for the following:

(i) $CH_2 = CH - C \equiv C - CH = CH_2$

(ii) $CH_2 - (CH_2)_3 - COOC(CH_3)_3$

(iii) $COH - (CH_2)_3 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_3$

(iv) $CH_3 - \underset{\displaystyle OH}{\underset{|}{CH}} - CH_2 - COOH$

(iv) $CH_3 - CO - CCl_3$

ANSWER:

(i) 1,5-Hexdiene-3-yne (ii) terbutyl pentanoate

(iii) 5-methylhexanal (iv)3-hydroxy butanoic acid

(v) 1,1,1-trichloro-2-propanon

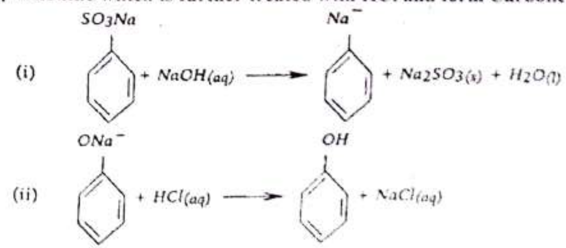
2. (xv) Give the preparation of the following:

* Carbolic acid from Benzene* Acetal from Methanal

*Ethanal from Acetylene * Diethyl ether from Ethyl alcohol

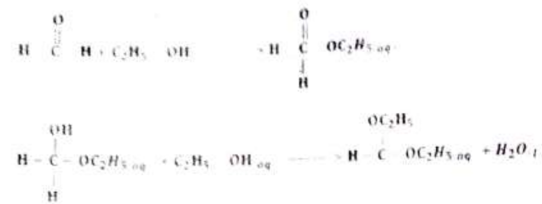
* Carbolic acid from Benzene

When Benzene sulphonate is fused with sodium hydroxide at 25 °C, it give sodium phenoxide which is further treated with HCl and form Carbolic acid



* Acetal from Methanal

When Methanal reacts with Ethyl Alcohol, it gives Acetal



*Ethanal from Acetylene

When Acetylene reacts with water in the presence of $HgSO_4$ and H_2SO_4 it gives Ethanal

$$(ii)\ H-C\equiv C-H + H_2O \xrightarrow{HgSO_4,\ H_2SO_4} \begin{bmatrix} CH = CH \\ \quad\ OH \end{bmatrix} \xrightarrow{\text{Rearrangement}} CH_3C-HO$$

* Diethyl ether from Ethyl alcohol

When Ethyl Alcohol is treated with concentrated Sulphuric acid at 140oC, then Diethyl ether is formed

$$2C_2H_5-OH_{(aq)} + H_2SO_{4(conc)} \xrightarrow{140^\circ C} C_2H_5-O-C_2H_5 + H_2O$$

2. (xvi) Give a simple chemical test to distinguish between the following:

*Alkane and Alkene *Ethene and Ethyne *Alkyl halide and Alkane

*Aldehyde and ketone

*ALKANE AND ALKENE

Alkenes give addition reaction while alkanes do not give addition reaction

$$CH_2 = CH_2 + H_2 \xrightarrow[\Delta H]{Ni} > H_3C - CH_3$$

$$CH_3 - CH_3 + H_2 \xrightarrow[\Delta H]{Ni} > No\ Reaction$$

*ETHENE AND ETHYNE

Ethene gives ethylene Glycol as following while Ethyne does not forms glycol.

$$CH_2 = CH_2 + H_2O + [O] \xrightarrow{KMnO_4} \underset{}{H_2\overset{OH}{\overset{|}{C}}-\overset{OH}{\overset{|}{C}}H_2}$$

*ALKYL HALIDE AND ALKANE

When alkyl halides react with sodium metal in presence of dry ether they gives alkane (Wurtz reaction) while alkane do not give Wurtz reactions.

$$2C_2H_5 - I + 2Na \xrightarrow[Ether]{Dry} > C_2H_5 - C_2H_5 + 2NaI$$

*ALDEHYDE AND KETONE

Aldehyde give Tollen's reagent test positive forming silver mirror while Ketones are unable to react with Tollen's reagent

$$H-\overset{O}{\overset{||}{C}}-H + [Ag(NH_3)_2]OH \longrightarrow 2Ag + H-\overset{O}{\overset{||}{C}}-O^-NH_4^+ + H_2O + 3NH_3$$

SECTION 'C'

INORGANIC CHEMISTRY

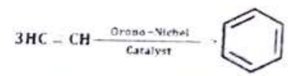
4. (b) Give complete and balanced equations of any five of the following:

1. $2K_2CrO_4 + H_2SO_4 \rightarrow K_2Cr_2O_7 + K_2SO_4 + H_2O$
2. $2CuSO_4 + 4KI \rightarrow 2CuI + 2K_2SO_4 + I_2$
3. $C_6H_{12}O_6 + \xrightarrow{Conc.H_2SO_4} 6C + 6H_2O$
4. $4Mg + 10HNO_3 \xrightarrow[cold]{dilute} 4Mg(NO_3)_2 + N_2O + 5H_2O$
5. $4FeO.Cr_2O_3 + 8K_2CO_3 + 7O_2 \rightarrow 2Fe_2O_3 + 8K_2CrO_4 + 8CO_2$
6. $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$

ORGANIC CHEMISTRY

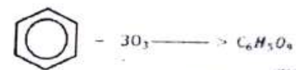
6. (c)Complete and balance the following equations:

Answer:



$$2HCHO + NaOH \longrightarrow > CH_3OH + HCOONa$$

$$CH_3COONa + NaOHCh \xrightarrow{CaO} > CH_4 + Na_2CO_3$$



$$C_6H_5O_9 \xrightarrow[Zn]{Hydrolysis} > 3\ \begin{matrix} CHO \\ | \\ CHO \end{matrix}$$

$$2CH_3-\overset{O}{\overset{||}{C}}-OH \xrightarrow[500^\circ C]{MnO_2} > H_3C-\overset{O}{\overset{||}{C}}-CH_3 + CO_2 + H_2O$$

CHEMISTRY 2017

SECTION 'B'

INORGANIC CHEMISTRY

2. (ii) Give scientific reasons for the following:

*** Alkali metals cannot be used in voltaic cell**

Answer: High negative values of standard electrode potential of Alkali metals indicate ease of oxidation. Since in voltaic cells water is used as solvent and Alkali metals are readily oxidized in water. Therefore they can not be used in voltaic cells.

*** Alkaline earth metals are harder than Alkali metals.**

Answer: Alkali metals and Alkaline earth metals form positively charged M+ and M++ ions respectively. Due to the greater charge attraction of metal ions and electron gas of Alkaline earth metal crystal, they are much harder than Alkali metals.

* Boric acid is soft and silky.

Boric acid is soft and silky, because the layers in Boric Acid Crystal structure slide over one another just like graphite.

* Anhydrous $CuSO_4$ is white while hydrated $CuSO_4$ is blue

Answer: In hydrated $CuSO_4$ the central metal atom copper is surrounded by the neutral atoms water and thus it gorms d-d transition and appears blue. Whereas the anhydrous cuso4 does not have water molecule in its compound it does not form d-d transition anddue to which it will be no colour generally it is said to be as whit and whereas hydrated $CuSO_4$ will be in blue colour.

2. (iii) Write the I.U.P.A.C. names of the following:

*$[Pt(NH_3)Cl_2]$ *$[Fe(CO)_4]$ *$Na_3[AlF_6]$ *$[CoBr_2(en)_2]Br$

ANSWER:

FORMULAE	I.U.P.A.C. NAMES
*$[Pt(NH_3)Cl_2]$	Tri ammine dichloro platinum (II)
*$[Fe(CO)_4]$	Tetra carbonyl iron (0)
*$Na_3[AlF_6]$	Sodium hexa fluoro Aluminate (III)
*$[CoBr_2(en)_2]Br$	Dibromo bis (ethylene diamine) Cobalt (III) bromide

2. (iv) Write the electronic configuration and identify the period, group and block of elements bearing the following atomic numbers : 17,24,29,49.

Answer:

Atomic No.	Electronic Configuration	Group	Period	Block
17	$1s^2, 2s^2, 2p^6, 3s^2, 3p^5$	VII-A	3	"p"
24	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^5$	VI-B	"4"	"d"
29	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^{10}$	I-B	"4"	"d"
49	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^2, 4d^{10}, 5p^1$	III-A	"5"	"p"

2. (v) Give complete and balanced equations for the following reactions

*Soda Ash heated with Silica

*Boric acid is reacted with Sodium Carbonate

*Litharge is heated with excess of air

*Aluminum is reacted with Sodium hydroxide

ANSWER:

(i) $Na_2CO_3 + SiO_2 \rightarrow Na_2SiO_3 + CO_2$

(ii) $4H_3BO_3 + Na_2CO_3 \rightarrow Na_2B_4O_7 + 6H_2O + CO_2$

(iii) $6PbO + O_2 \xrightarrow{450°C} 2Pb_3O_4$

(iv) $2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$

2. (viii) Refer the list of the given compounds:

Compound	A	B	C	D
Specific Name	Gypsum	Bleaching powder	Lunar Caustic	Potash Alum

* Formulae of "B" and "C"

Answer:

FORMULAE OF B AND C

Formula of B: $CaOCl_2$ Formula of C: $AgNO_3$

* The equation for the preparation of B

$$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$$

* Given only one common use of "D"

Potash Alum is used for purification of drinking water as a chemical flocculent.

* Given equation for the reaction of heating 'A' at 100°C

$$CaOCl_2 + 2HCl \longrightarrow CaCl_2 + H_2O + Cl_2$$

ORGANIC CHEMISTRY

2. (xi) Write equations for the following reactions:

*Cannizaro reaction *William Son Synthesis

*Fermentation of Molasses *Esterification

*CANNIZARO REACTION

$$H-CHO_{(aq)} + H-CHO_{(aq)} \rightarrow CH_3OH + HCOO^-Na^+$$

*WILLIAM SON SYNTHESIS

$$2CH_3-CH_2-OH_{(aq)} + 2Na_{(s)} \rightarrow 2CH_3-C_2-O.Na + H_{2(g)}$$

$CH_3-CH_2-O^-.Na^+ + CH_3-CH_2^--Br^- \rightarrow CH_3-CH_2-O-CH_2-CH_{3(aq)} + NaBr_{(aq)}$

$CH_3-CH_2-OH + H^+O^--CH_2-CH_3 \xrightarrow{H_2SO_4} CH_3-CH_2-O-CH-2-CH_{3(aq)} + H_2O_{(l)}$

*FERMENTATION OF MOLASSES

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{Sucrose\ Inyeast} C_6H_{12}O_6 + C_6H_{12}O_6$$

$$C_6H_{12}O_{6(s)} \xrightarrow{Zymase\ in\ yeast} 2C_2H_5-OH + 2CO_2$$

*ESTERIFICATION

$$CH_3-\overset{\overset{O}{||}}{C}-OH + C_2H_5-OH + H_2SO_{4(aq)} \xrightarrow{Heat} CH_3-\overset{\overset{O}{||}}{C}-O-C_2H_5 + H_2O$$

2. (xii) Write I.U.P.A.C. Name of the following

* $CH_2=CH-CH_2-COOH$ * $C_2H_5-(CH_2)_3-COCH(CH_3)_2$

* $CH_3-C(CH_3)_2-COCH(CH_3)_2$

* $CH_3-CH_2-OC(CH_3)_2C_2H_5$

ANSWER:

(i) 3- butenoi acid (ii) Isopropyl hexanoate

(iii) ,2,4-tri methyl pentan-3-one (iv)2-ethoxy-2-methyl butane

2. (xvi) Distinguish by simple chemical test only two of the following:

*Ethyne and Ethene

(See Question No. 2(xiv), 2018)

*Ethane and Ethene

Ethene give addition reaction while Ethane do not give addition reaction

$$CH_2=CH_2 + H_2 \xrightarrow[\Delta H]{Ni} H_3C-CH_3$$

$$CH_3-CH_3 + H_2 \xrightarrow[\Delta H]{Ni} No\ Reaction$$

*Alkyl Halide and Alkane

(See Question No. 2(xiv), 2018)

*Formaldehyde and Acetone.

Formaldehyde gives silver mirror test with Febling's Solution while Acetone do not gives this reaction.

$$H-\overset{\overset{O}{\|}}{C}-H + 2Cu(OH)_{2(aq)} + NaOH_{(aq)} \longrightarrow H-\overset{\overset{O}{\|}}{C}-Na + Cu_2O + 3H_2O$$

SECTION 'C'

INORGANIC CHEMISTRY

4. (b) Give complete and balanced equation(Any five)

Answer:

(i) $2CaOCl_2 + CO_2 + H_2O \rightarrow CaCO_3 + CaCl_2 + 2HOCl$

(ii) $2Al + Fe_2O_3 \rightarrow Al_2O_3 + 2Fe$

(iii) $K_2Cr_2O_7 + 4\,KCl + 6H_2SO_4 \rightarrow 6KHSO_4 + 2CrO_2Cl_2 + 3H_2O$

(iv) $P + 5HNO_3 \rightarrow H_3PO_4 + 5NO_2 + H_2O$

(v) $Sb_2S_3 + 6HCl \rightarrow 2SbCl_3 + 3H_2S$

(vi) $4Zn + 5H_2SO_4 \rightarrow 4ZnSO_4 + H_2S + 4H_2O$

ORGANIC CHEMISTRY

6. (b) How the following conversing carried out

(i) Acetone into ter.butyl alcohol

$$CH_3-MgI + CH_3-CO-CH_3 \longrightarrow CH_3-\underset{CH_3}{\overset{CH_3}{\overset{|}{\underset{|}{C}}}}-O-Mg-I$$

$$CH_3-\underset{CH_3}{\overset{CH_3}{\overset{|}{\underset{|}{C}}}}-O-Mg-I \xrightarrow[\text{Dil.HCl}]{H-OH} CH_3-C-OH-Mg$$

(ii) Ethanol into chloro ethane

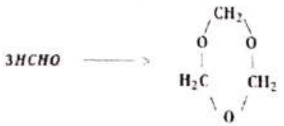
$$C_2H_5OH + HCl \xrightarrow{ZnCl_2} C_2H_5Cl + H_2O$$

(iii) Diethyl ether into Ethyl iodide

$$C_2H_5-O-C_2H_5 + HI \longrightarrow C_2H_5OH + C_2H_5I$$

$$C_2H_5OH + HI \longrightarrow C_2H_5IO + H_2O$$

(iv) Formaldehyde into meta formaldehyde



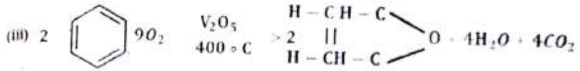
(v) Acetyl chloride to Acetic anhydride.

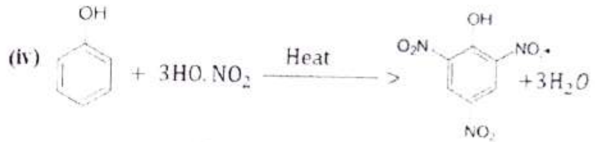
6. (c) Complete the following reactions:

ANSWER:

(i) $CH_2=CH-CH_3 + HBr \longrightarrow CH_3-\underset{Br}{\underset{|}{CH}}-CH_3$

(ii) $H-C\equiv C-H + H_2O \xrightarrow{HgSO_4,\ H_2SO_4} \begin{bmatrix} CH=CH \\ \quad | \\ \quad OH \end{bmatrix} \xrightarrow{\text{Rearrangement}} CH_3C-HO$





CHEMISTRY 2016

SECTION 'B'

INORGANIC CHEMISTRY

2. (i) Write I.U.P.A.C. names of the following:

*[Co(NH₃)₃Cl] *[Cr(en)₃]Cl₃

*Na₃[Cu(OH)₂(NO₃)₂] *Na₂[Fe(CN)₅NO]

ANSWER:

FORMULAE	I.U.P.A.C. NAMES
*[Co(NH₃)₃Cl]	Triamine cobalt (III) chloride
*[Cr(en)₃]Cl₃	Tris (Ethylenediomine) chromium (III) chloride
*Na₃[Cu(OH)₂(NO₃)₂]	Sodium dihydroxo dinitrato Cuperate (I)
*Na₂[Fe(CN)₅NO]	Sodium pentacyeno nitrosyl Ferrate (II).

2. (iii) Give scientific reasons for the following:

*Zinc hydroxide is soluble in excess of Sodium hydroxide solution.

AnswerWhen excess NaOH reacts with insoluble Zinc hydroxide its yield complex Tetrahydroxozincate(II) ion which is soluble in water

*Alkali and alkaline earth metals from only +1 and +2 ions respectively

Answer: Because alkali metal have 1 electron in their outermost shell so it will donate this one electron to become stable and thus it forms 1 ion.

In the same way, Alkali earth metal have 2 electrons in their outermost shell so it will donate these two electrons to become stable and thus forming 2 ions.

*Atomic size of Sulphur is bigger than that of Oxygen

Answer: Oxygen and slupher are belongs to same group in periodic table. But sulpher placed in 3rd period while oxygen in 2nd. It means that sulpher has 3 orbits of electrons while oxygen has only 2 orbits. Due to this reason Sulphur is bigger than Oxygen

* Plastic Sulphur is elastic.

Answer: Plastic sulphur is an amorphous form of sulphur when liquid (molten) sulphur is poured in ice water the immediate freezing of material is responsible for irregular arrangement of atoms but a coils like structure is formed. The coiling and uncoiling of the structure is responsible for stretching or elastic nature of sulphur.

2. (vi) Write equations for the following reactions:

* Conc. Suphuric acid with Oxalic acid *Nitric acid with Sulphur
*Fe^{+3} wihH_2S * Blue stone treated with KI

ANSWER:

(i) $(COOH)_2 \xrightarrow{Conc.H_2SO_4} CO_2 + CO + H_2O$

(ii) $S + 6HNO_3 \rightarrow H_2SO_4 + 6NO_2 + 2H_2O$

(iii) $2Fe^{3+} + H_2S \rightarrow 2Fe^{2+} + S + 2H^+$

(iv) $2CUSO_4 + 4KI \rightarrow Cu_2I_2 + 2K_2SO_4 + I_2$

2. (vii)(a) Write the chemical formulae of the following:

*Suhaga *Alunite *Murda sang *Chromite ore

Answer

SUBSTANCE	CHEMICAL FORMULAE
*Suhaga	$Na_2[B_4O_5(OH)_4]\cdot 8H_2O$
*Alunite	$KAl_3(SO_4)_2(OH)_6$
*Murda sang	PbO
*Chromite ore	$FeCr_2O_4$

ORGANIC CHEMISTRY

2. (xi) Give chemical test to distinguish between the following:

*ALDEHYDE AND KETONE

(See Question No. 2(xiv), 2018)

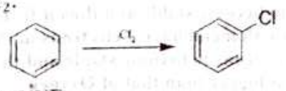
*PARAFFIN AND OLEFIN

Olefins decolourized $KMnO_4$ while paraffins are unable to decolourize $KMnO_4$

$$CH_2 = CH_2 + H - OH + [O] \xrightarrow{KMnO_4} CH_2(OH) - CH_2(OH)$$

*n-HEXANE AND BENZENE

Benzene gives addition reaction with Cl_2 at 50 °C and 400 atm produce Hexachloro cyclo hexane but Hexane is a saturated hydrocarbon, it does not give addition reactionCl_2.



*1-BUTYNE AND 2-BUTYNE

When 1-Butyne is treated with $AgNO_3$ it gives white precipitate while 2-Butyne does not react with $AgNO_3$.

$CH_3.CH_2.C \equiv CH + AgNO_3 + 2NH_4OH \longrightarrow CH_3CH_2CCAg + 2NH_4NO_3 + H_2O$

$CH_3.C \equiv C.CH_3 + AgNO_3 + 2NH_4OH \longrightarrow$ No Reaction

2. (xiii) What happens when?

* Ethylene react with cold aqueous $KMnO_4$ solution.

When Ethylene react with aqueous $KMnO_4$ it yields Ethylene Glycol

$$CH_2 = CH_{2(g)} + H_2O_{(l)} + [OH] \xrightarrow[Cold]{KMnO_4} OH - CH_2 - CH_2 - OH$$

* Mehtanol reacts with steam

When Methanol react with steam, it yields carbon dioxide and Hydrogen

$$CH_3 - OH_{(l)} + H_2O_{(g)} \xrightarrow[Cold]{250\ °C} CO_{2(g)} + 3H_{2(g)}$$

*2-Propanone is oxidized in presence of $K_2Cr_2O_7$ and conc. H_2SO_4.

When 2-propanon is oxidized in presence of $K_2Cr_2O_7$ and conc. H_2SO_4 it yields acetic acid.

$$CH_3 - CO - CH_{3(aq)} + 4[O] \xrightarrow[H_2SO_4]{K_2Cr_2O_7} CH_3 - CO - OH_{(aq)} + H_2O_{(l)}$$

* Ethyl chloride reacts with Sodium metal.

When Ethyl chloride reacts with Sodium metal in the presence of ether. It yields Butane.

$$2(CH_3 - CH_2 - Cl)_{(aq)} + 2Na_{(s)} \xrightarrow[Solvent]{Ether} C_4H_{10(aq)} + 2NaCl_{(s)}$$

SECTION 'C'

Inorganic Chemistry and one from Organic Chemistry.

INORGANIC CHEMISTRY

b) Complete and balance the following equation:

Answer:

(i) $K_2Cr_2O_7 + 2H_2SO_4 \rightarrow 2KHSO_4 + 2CrO_3 + H_2O$

(ii) $4KI + K_2Cr_2O_7 + 6H_2SO_4 \rightarrow 6KHSO_4 + 2CrO_3 + H_2O$

(iii) $6NaOH + 3Cl_2 \rightarrow 5NaCl + NaClO_3 + 3H_2O$

(iv) $P + 5HNO_3 \xrightarrow[dil.]{cold} H_3PO_4 + 5NO_2 + H_2O$

(v)) $2KMnO_4 + Cl_2 \rightarrow 2KMnO_4 + 2KCl$

ORGANIC CHEMISTRY

5. (a) Give the equations of the following:

*Reaction of Ethanal with Grignard reagent

$$(CH_3)_2 - C^{+\delta} = O^{-\delta} + CH_3 - Mg - Cl \rightarrow (CH_3)_2 - HC - O^- Mg^+ Cl \xrightarrow{H_3^+O} (CH_3)_2 - HC - OH + Cl - Mg - OH$$

* Decomposition of Acetic acid in the presence of MnO_2

$$2CH_3 - C - OH_{(g)} \xrightarrow[MnO_2]{500\ °C} CH_2 - C + CH_3 + CO_{2(g)} + H_2O_{(l)}$$

* Reaction of Methanal with caustic soda.

$$H - CHO_{(aq)} + H - CHO_{(aq)} + NaOH \rightarrow CH_3 - OH_{(l)} + HCO^- ONa^+_{aq}$$

* Reaction of Chloromethane with Sodium ethoxide.

$$CH_3 - CH_2 - O.Na^+_{aq} + CH_3 - CH_2 - Cl_{aq} \longrightarrow CH_3 - CH_2 - O - CH_{3(l)} + NaCl_{(aq)}$$

5. (b) Give the preparation of the following:

* 2-Bromo propane from 1-Bromo propane

When 1-Bromo propane reacts with potassium hydroxide then 2-bromo-propane is formed

$$CH_3 - CH_2 - CH_2 - Br_{aq} + KOH \longrightarrow CH_3 - CH = CH_{2(aq)} + KBr_{(aq)} + H_2O_{(l)}$$

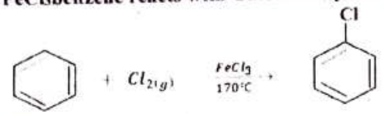
$$CH_3 - CH = CH_{2(aq)} + HBr_{(aq)} \rightarrow (CH_3)_2 - CH - Br$$

* Phenol from Benzene

Phenol is obtain from Benzene by the following process

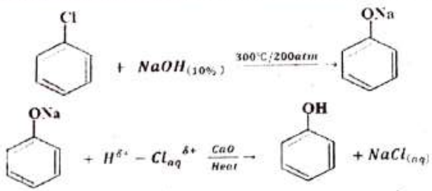
FORMATION OF CHLOROBENZENE

in the presence of $FeCl_3$benzene reacts with Chlorine it yield Chlorobenzene



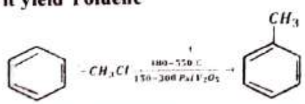
FORMATION OF PHENOL

When Chlorobenzene is heated with 10% Concentration NaOH it gives Phenol

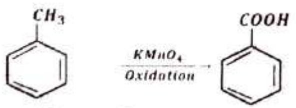


* Benzoic acid from Benzene

When Benzene reacts with Methyl chloride in the presence of V_2O_5 at 480-550°C and 150-300 Psi it yield Toluene



When Toluene is Oxidized in the presence of $KMNO_4$ it produced Benzoic Acid.



*Isopropyl alcohol from 1-Propanol

When 1-Propanol treated with Suphuric acid it gives Isopropyl alcohol

$$CH_3 - CH_2 - CH_2 - OH_{(l)} + H_2SO_{4(aq)} \rightarrow CH_3 - CH = CH_{2(g)} + H_2O$$

$$CH_3 - CH = CH_{2(g)} + H_2O \rightarrow CH_3 - \underset{OH}{\underset{|}{CH}} - CH_3$$

6. (a) Write the I.U.P.A.C. names of the following:

(i) $(C_6H_5)_3CBr$ (ii) $C_2H_5OCH(CH_3)_2$

(iii) $H_2C = CH - \overset{Cl}{\overset{|}{CH}} - CH_2 - \overset{CH_3}{\overset{|}{CH}} - C - CH$

(iv) $(CH_3)_2CH - \overset{O}{\overset{\|}{C}} - C(CH_3)_3$

(v) $CH_3 - \overset{O}{\overset{\|}{C}} - CH_2 - COOH$

ANSWER:

(i) Triphenyl bromo methane (ii)2-ethoxy propane

(iii) 3-chloro-5-methyl-1-heptene-6-yne

(iv) 2,2,4-trimethyl-3-pentanone (v) 3-oxo-butanoic acid

CHEMISTRY 2015

SECTION 'B'

INORGANIC CHEMISTRY

ii) Give scientific reasons for the following:

* Hydrogen exhibits +1 and -1 oxidation states in its compounds.

ANSWER: " Hydrogen exhibits +1 and -1 oxidation states in its compounds. Due to half-filled valence orbital hydrogen exhibit +1 & -1 oxidation states by either gain one electron to complete its orbit forming H- or by losing its valence electron forming H+.

* Alkali metals are powerful reducing agents.

ANSWER: Alkali metals have one electron in their valence shell. They readily loose one electron to attain stable electronic configuration and hence by losing electron they oxidize themselves and reduce other compounds.

* SO_3 is dissolved in H_2SO_4 but not in water.

ANSWER: In the preparation of sulphuric acid. SO_3 is first dissolved in H_2SO_4 to form oleum & not in water. But the reaction with water will be uncontrollable & it creates a fog of sulphuric acid.

* Transition elements show variable oxidation states

ANSWER: Transition elements show variable oxidation states due to the participation of outer ns and inner (n–1)d-electrons in bonding. Except scandium, the most common oxidation state shown by the elements of first transition series is +2. This oxidation state arises from the loss of two 4s electrons. This means that after scandium, d-orbitals become more stable than the s-orbital.

2. (iii) Complete the following reactions and give I.U.P.A.C. names of the products.

* $AgOH + NH_4ON \rightarrow$ * $Zn(OH)_4 + NaOH \rightarrow$

Answer: $AgOH + NH_4ON \rightarrow [Ag(NH_3)_2]OH + H_2O$

Diamine Silver(I)Hydroxide

$Zn(OH)_4 + NaOH \rightarrow Na_2[Zn(OH)_4]^-$

Sodium tetrahydro oxozinate (II)

2. (v) Give chemical equations for the following:

* Reaction of Aluminium with conc. H_2SO_4.

* Reaction of Sodium carbonate with Silica
* Reaction of Caustic soda with Ammonium Sulphate.
* Reaction of bleaching powder with atmospheric carbon dioxide and moisture.

ANSWER:

(i) $2Al + 6H_2SO_4 \xrightarrow{conc} Al_2(SO_4)_3 + 6H_2O + 3SO_2$

(ii) $Na_2CO_3 + SiO_2 \rightarrow Na_2SiO_3 + CO_2$

(iii) $2NaOH + (NH_4)_2SO_4 \rightarrow Na_2SO_4 + 2NH_3 + 2H_2O$

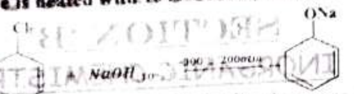
(iv) $2CaOCl_2 + CO_2 + H_2O \rightarrow CaCO_3 + CaCl_2 + 2HOCl$

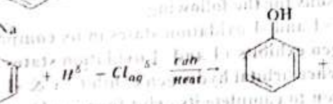
ORGANIC CHEMISTRY

2. (xiv) Write equation for the preparation of the following:

* Phenol from Chlorobenzene

When Chlorobenzene is heated with 10% Concentration NaOH it gives Phenol





* Ethyl ethanoate from Ethanol

$$CH_3-CH_2-OH_{(l)} + HO-\overset{O}{\overset{||}{C}}-CH_{3(aq)} \rightarrow C_2H_5-O-\overset{O}{\overset{||}{C}}-CH_{3(l)} + H_2O_{(l)}$$

* Water gas from methane

$$\left[CO_{(g)} + H_{2(g)} \xrightarrow[400-450^\circ C/200atm]{ZnO/Cr_2O_3} CH_3OH\right]$$

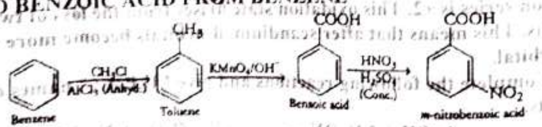
* Ethene from Ethyl bromide

$$CH_3-CH_2-Br_{(l)} + KOH_{(aq)} \xrightarrow{Alcoholic} CH_2=CH_{2(g)} + KBr_{(s)} + H_2O_{(l)}$$

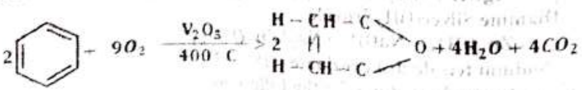
2. (xvi) Write the stepwise reactions for the following preparations.

* m-Nitro benzoic acid from Benzene
* Glyoxal from Benzene

*M-NITRO BENZOIC ACID FROM BENZENE



GLYOXAL FROM BENZENE



SECTION 'C'

INORGANIC CHEMISTRY

3. (c) Complete the following equations:

Answer:

(i) $4Zn + 5H_2SO_4 \xrightarrow{90\%} 4ZnSO_4 + H_2S + 4H_2O$

(ii) $AgBr + 2Na_2S_2O_3 \rightarrow Na_3[Ag(S_2O_3)_2] + NaBr$

(iii) $P + 5HNO_3 \rightarrow H_3PO_4 + 5NO_2 + H_2O$

(iv) $Ca_2B_6O_{11} + 2Na_2CO_3 \rightarrow 2CaCO_3 + Na_2B_4O_7 + 2NaBO_2$

(v) $K_2Cr_2O_7 + 6FeSO_4 + 7H_2SO_4 \rightarrow 3Fe_2(SO_4)_3 + K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O$

ORGANIC CHEMISTRY

5. (c) What happens when:

* **Acetone is treated with I_2 in the presence of Sodium carbonate.**

When Acetone is treated with I_2 in the presence of Sodium carbonate, it yields yellow precipitates of **iodoform**

$$CH_3-CO-CH_{3(aq)} + 3I_{2(s)} 2Na_2CO_{3(s)} \xrightarrow{Heat} CH_3CO-ONa_{(s)} + 3NaI_{(s)} + 2CO_{2(g)} + H_2O_{(l)}$$

*Methyl iodide is treated with Sodium metal.

Methyl iodide is treated with Sodium metal, it yields Ethane.

$$H_3C-I + 2Na_{(s)} + I-CH_{3(aq)} \xrightarrow{ROR} 2NaI_{(s)} + 2CH_{3(g)}$$

$$2CH_{3(g)} \longrightarrow C_2H_{6(g)}$$

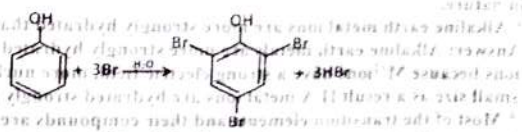
* Excess Ethanol is treated with Conc. H_2SO_4.

When Excess Ethanol is treated with Conc. H_2SO_4 it yields Diethyl ether

$$2C_2H_5OH \xrightarrow[140^\circ C]{H_2SO_4} C_2H_5-O-C_2H_5 + H_2O$$

***Phenol is treated with Bromine water.**

When Phenol reacts rapidly with bromine water and produces 2,4,6-tri bromo phenol



6. (c) Write I.U.P.A.C. names of the following

(i) $(CH_3)_3C-CH_2-CO-CH(CH_3)_2$

(ii) $CH_3-CH_2-\overset{O}{\overset{||}{C}}-CH_2-COOH$

$$(iii)\ CH_3-(CH_2)_3-\overset{\overset{O}{\|}}{C}-O-C(CH_3)_3$$

$$(iv)\ CH_3-O-\underset{\underset{OC_2H_5}{|}}{CH}-CH_2-\underset{\underset{OC_6H_5}{|}}{CH_2}$$

$$(v)\ CH_2=CH-C\equiv C-CH=CH_2$$

ANSWER:
(i) 2,2,5- trimethyl hexan-3-one
(ii) 3-oxo-pentanoic acid
(iii) Terbutl pentanoate
(iv) 1-ethoxy-1-methoxy-3-phenoxy propane
(v) 1,5-hexadien-3-yne

CHEMISTRY 2014

SECTION 'B'

INORGANIC CHEMISTRY

2. (ii) Give scientific reasons for any four:

*** NH_3 and H_2O acts as ligands but NH_4^+ and H_3O^+ do not.**

Answer: In H_2O and NH_3, there is a lone pair of electron which is available for donation. Therefore they act as ligands while in a case of H_2O^+ and NH_4^+ Ion pair of electron is not available-in fact it is consumed in making of coordinate covalent bond.

*** Graphite conducts electricity, whereas diamond does not.**

Answer:In diamond the carbon atoms are bonded to four other atoms. Whereas, in graphite each carbon atom is only bonded to three other atoms. Therefore, there are delocalised (free) electrons in graphite, which can move and carry a charge so graphite conducts electricity.

*** B_2O_3 is acidic while Al_2O_3 is amphoteric**

Answer: Boron trioxide i.e., B_2O_3 is an anhydride of boric acid I.e., $B(OH)_3$ or H_3BO_3. Hence, it is said to be acidic. But, in the case of aluminium, since it is a metalloid(neither exactly a metal nor a non metal) and has decreased non metallic properties compared to boron, the oxide of aluminium is amphoteric in nature.

*** Alkaline earth metal ions are more strongly hydrated than alkali metal ions.**

Answer: Alkaline earth metals are more strongly hydrated than alkali metals ions because M^{+2}ions have a strong electric field, more nuclear charge and small size as a result II A metal ions are hydrated strongly than IA ions.

*** Most of the transition elements and their compounds are paramagnetic.**

Answer: The transition metals and their compounds show magnetic properties. Many compounds of the transition metals are paramagnetic because of unpaired electron spins in the atom.

2. (iv)Give equation for the following reactions:
* Chrome yellow with Caustic soda.
* Electrolytic oxidation of Potassium manganite by water
*Potassium dichromate with Conc. H_2SO_4.
* Chromium oxide with KOH and Bromine water.

ANSWER:
(i) $2PbCrO_4 + 2NaOH \rightarrow Na_2CrO_4 + Pb_2CrO_5 + H_2O$
(ii) $2K_2MnO_4 + 2H_2O \rightarrow 2KMnO_4 + 2KOH + H_2O$
(iii) $K2Cr_2O_7 + 2H_2SO_4 \rightarrow 2KHSO_4 + 2CrO_3 + H_2O$
(iv) $Cr_2O_3 + 10\ KOH + 3Br_2 \rightarrow 2K_2CrO_4 + 6KBr + 5H_2O$

2. (v) (a) Write I.U.P.A.C. names of the following:
*$[Fe(NO_2)_6]^{-3}$ *$[Cr(en)_2Cl_2]Cl$

ANSWER:
*$[Fe(NO_2)_6]^{-3}$ Hexa nitro ferrate (III) ion
*$[Cr(en)_2Cl_2]Cl$ Bis (ethylene diamine) dichlorochromium (III) chloride

2. (vi)(a) Write the electronic configuration of Zn (Z=30) , group, period and block

Answer:

Electronic Configuration	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}$
Group	II-B
Period	4
Block	d

2. (vii)(b)Write the formulae for the following:
*Alunite *Hypo *Magnesite *Fluorspar

Answer:

SUBSTANCE	CHEMICAL FORMULAE
Alunite	$KAl_3(SO_4)_2(OH)_6$
*Hypo	$Na_2S_2O_2.5H_2O$
*Magnesite	$MgCO_3$
*Fluorspar	CaF_2

ORGANIC CHEMISTRY

2. (xii) Give equations for the following reactions:

*** methanol with steam**

$$CH_3-OH_{(l)} + H_2O_{(l)} \xrightarrow{250\ ^\circ C} CO_2 + 3H_{2(g)}$$

*** Glucose with Tollen's Reagent**

$$C_6H_{12}O_{6(s)} + 2[Ag(NH_3)_2]OH_{(aq)} \longrightarrow 2Ag_{(s)} + \begin{array}{c} CH_2OH \\ | \\ (CHOH)_4 \\ | \\ CH_2OH \end{array}$$

*** Acetyl chloride with Sodium ethanoate**

$$CH_3CNa_{(s)} + CH_3-\overset{\overset{O}{\|}}{C}-Cl_{(aq)} \longrightarrow CH_3-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-CH_3 + NaCl_{(s)}$$

*** Ethanol with Grignard Reagent**

$$C_2H_5OH_{(l)} + CH_3-Mg-Cl \longrightarrow CH_3-CH_3 + OHCH_3-Mg-Cl$$

SECTION 'C'

INORGANIC CHEMISTRY

3. (c) Complete the following equations:

Answer:

(i) $P + 5HNO_3 \rightarrow H_3PO_4 + 5NO_2 + H_2O$

(ii) $4Zn + HNO_3 \xrightarrow{very\ dil.} 4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$

(iii) $H_2S + H_2SO_4 \rightarrow SO_2 + S + 2H_2O$

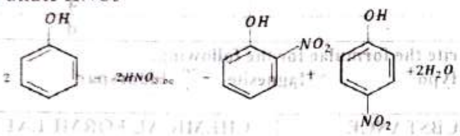
(iv) $2CaOCl_2 + CO_2 + H_2O \rightarrow CaCO_3 + CaCl_2 + 2HOCl$

(v) $AgBr + 2\ Na_2S_2O_3 \rightarrow Na_3[Ag(S_2O_3)_3] + NaBr$

ORGANIC CHEMISTRY

5. (c) Give equation for the following reactions:

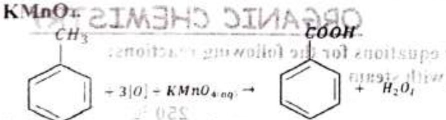
* **Phenol with dilute HNO₃**



* **Ethanal with Sodium and Ethanol**

$$CH_3 - \overset{\overset{\displaystyle O}{||}}{C} - H + 2[H] \xrightarrow{Na\quad C_2H_5OH} CH_3 - CH_2 - OH$$

* **Toluene with KMnO₄**



* **2-Propanone with Methyl magnesium chloride.**

$$CH_3 - \overset{\overset{\displaystyle O}{||}}{C} - CH_3 - \quad CH_3 - Cl_{(aq)} + H_2O_{(l)} \longrightarrow CH_3 - C - OH + OH - Mg - Cl$$

* **Acetylene with cold KMnO₄.**

$$CH \equiv CH + H_2O + 3[O] \xrightarrow[Cold]{KMnO_4} 2HCOOH$$

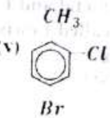
6. (c) Write I.U.P.A.C. names of the following;

(i) $CH_2 - C - CH \equiv CH$ (with CH_2 below C)

(ii) $CH_3 - \overset{\overset{\displaystyle O}{||}}{C} - O - C(CH_3)_3$

(iii) $CH_3 - CH = CH - \underset{OH}{CH} - \underset{CH_3}{CH} - CH_3$

(iv) $CH_3 - CH_2 - O - C(CH_3)_3$

(v) 

ANSWER:

(i) 2-methyl-1-buten-3-yne **(ii) ter butyl ethanoate**

(iii) 2 -methyl-4-hexen-3-ol **(iv) 2-methyl-2-ethoxy propane**

(v) 4-bromo-2-chloro toluene

CHEMISTRY 2013

SECTION 'B'

INORGANIC CHEMISTRY

2. (i) (a) Give the valence shell electronic configuration of the following groups:

*IIA and IB *IVA and VIB

Group II-A: Electronic configuration of valence shell is ns^2

Group I-B: Electronic configuration of valence shell is ,ns^1,$(n-1)d^{10}$

Group V-A: Electronic configuration of outer valence is ns^2 or np^3

Group V-B: Electronic configuration of valence shell is ns^2,$(n-1)d^3$

2. (iii) Give equations for the following:

* Reaction of Ferric oxide with Aluminium.

* Reaction of Litharge with Sodium chloride

* Reaction of Benzene with Sulphuric acid

* Reaction of Sodium dichromate with Potassium chloride

ANSWER:

(i) $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$

(ii) $5PbO + H_2O + 2NaCl \rightarrow 2NaOH + PbCl_2.4PbO$

(iii) $C_6H_6 + HO - SO_3H \rightarrow C_6H_5 - SO_3H + H_2O$

(iv) $Na_2Cr_2O_7 + 2KCl \rightarrow 2NaCl + K_2C_2O_7$

2. (v) Give reasons:

*** The elements of a group in the periodic table have the same valence shell electronic configuration.**

Answer: Since the properties of element have a periodic function of their atomic no (Z). The electronic configuration varies with increasing atomic no. in periodic manner. Hence in a groups, elements shows same valence shell electronic configuration.

*** Alkali metals cannot be used in voltaic cell.**

Answer: High negative values of standard electrode potential of Alkali metals indicate ease of oxidation. Since in voltaic cells water is used as solvent and Alkali metals are readily oxidized in water. Therefore they can not be used in voltaic cells.

*** In d block elements, 4s orbital are filled prior to 3d orbitals but 4s elections are lost first in ionization.**

Answer: 4s orbitals has less energy than 3d, So it filled prior to 3d. But 4s orbit is outer most orbital that's why electron lost first from 4s orbit in Ionization.

***Ligands are generally called Lewis bases.**

ANSWER: ligands donates electron to transition metal and Lewis base also electron donner. Therefore ligands are generally called Lewis bases.

2. (vii) (a) Write I.U.P.A.C. names of the following:

*$NH_4[Cr(SCN)_4(NH_3)_2]$ *$Na_3[Fe(CN)_5NO]$

ANSWER:

*$NH_4[Cr(SCN)_4(NH_3)_2]$

Ammonium diamminetetrathiocyanato-S Chromate (III)

*$Na_3[Fe(CN)_5NO]$

Sodium pentacyanonitrosyl ferrate (II)

2. (vii) (b)Write the chemical formulae of the following:

*Tincal *Lead sesquioxide *Stibnite *Carnallite

Answer:

SUBSTANCE	CHEMICAL FORMULAE
Tincal	$Na_2B_4O_7.10H_2O$
*Lead sesquioxide	$O_3Pb_2^{-4}$
*Stibnite	Sb_2S_3
*Carnallite	$KMgCl_3\cdot 6(H_2O)$

ORGANIC CHEMISTRY

2. (xi) (b) Writhe equation for the reactions of Alcoholic KOH with:

*2-Bromo propane *2-Bromo-2-methyl propane

2-BROMO PROPANE

$$CH_3-\underset{}{\overset{Br}{\overset{|}{CH}}}-CH_3+KOH \longrightarrow CH_3-CH=CH_2+KBr+H_2O$$

2-BROMO-2-METHYL PROPANE

$$CH_3-\overset{CH_3}{\overset{|}{\underset{Br}{\underset{|}{C}}}}-CH_3+2Na+CH_3-\overset{CH_3}{\overset{|}{\underset{Br}{\underset{|}{C}}}}-CH_3 \longrightarrow CH_3-\overset{CH_3}{\overset{|}{CH}}-\overset{CH_3}{\overset{|}{CH}}-CH_3 \quad 2NaBr$$

2. (xii) Name the following compounds by I.U.P.A.C. system:

*$(CH_3)_2C.Br.CHO$ *$CH_3(CH_2)_3COOC(CH_3)_3$

*$(CH_3)_3C.C.OH$ *$(CH_3)_2CHCOC(CH_3)_3$

ANSWER:

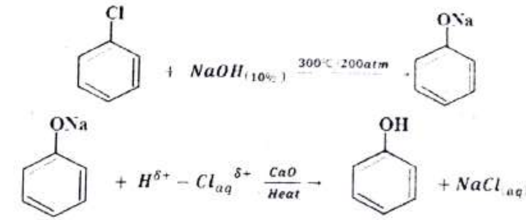
(i) 2-bromo-2-methyl propanal (ii) terbutyl pentanoate

(iii) 2-methyl-2-propanol (iv) 2,2,4-trimethyl pentan-3-one

2. (xiv) How will you prepare the following?

*** Carbolic acid from Benzene**

When Chlorobenzene is heated with 10% Concentration NaOH it gives Carbolic acid or Phenol



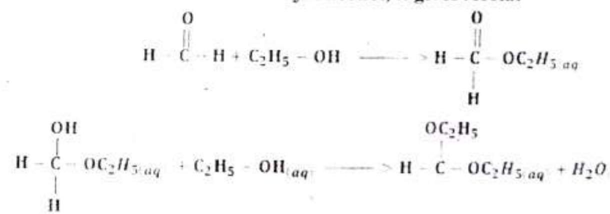
*** Acetaldehyde from Acetylene**

Acetylene reacts with water in presence of Mercuric Sulphate and Sulphuric acid at 75°C o produce Acetaldehyde

$$H-C\equiv C-H+H_2O \xrightarrow[75^\circ C]{HgSO_4/H_2SO_4} > CH_3CHO$$

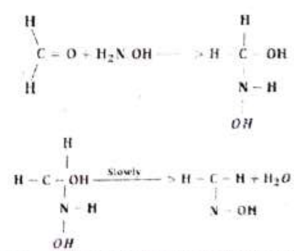
*** Acetal from Methanal**

When Methanal reacts with Ethyl Alcohol, it gives Acetal

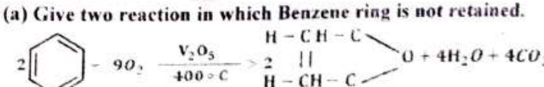


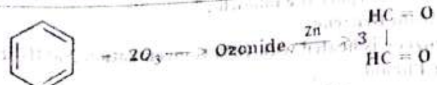
***An Oxime from Formaldehyde**

Formaldehyde adds hydroxalamine (NH_2OH) to form a stable product called "Oxime".



2. (xvi) (a) Give two reaction in which Benzene ring is not retained.





SECTION 'C'

INORGANIC CHEMISTRY

3. (b) Complete the following Equation

Answer:

(i) $4FeO.Cr_2O_3 + 8K_2CO_3 + 7O_2 \rightarrow 2Fe_2O_3 + 8K_2CrO_4 + *CO_2$

(ii) $2KMnO_4 + Cl_2 \rightarrow 2KMnO_4 + 2KCl$

(iii) $2CuSO_4 + 4KI \rightarrow Cu_2I_2 + 2K_2SO_4 + I_2$

(iv) $Zn + HNO_3 \xrightarrow{90\%} 4ZnSO_4 + H_2S + 4H_2O$

(v) $MgCO_3 + H_2SO_4 \rightarrow MgSO_4 + CO_2 + H_2O$

ORGANIC CHEMISTRY

5. (b) Prepare the following compounds from CH_3Br:

* Methanol *Methyl thanoate *Methoxy ethane *Amino methane

*Methl mercaptan (Methyl thioalcohol)

METHANOL

$$CH_3Br + NaOH \longrightarrow CH_3OH + NaBr$$

METHYLETHANOATE

$$CH_3Br + CH_3COONa \longrightarrow CH_3COOCH_3 + NaBr$$

METHOXYETHANE

$$CH_3Br + CH_3ONa \longrightarrow H_3 - OH_3 - CH_3 + NaBr$$

AMINO METHANE

$$CH_3Br + NH_3Na \longrightarrow CH_3NH_2 + NaBr$$

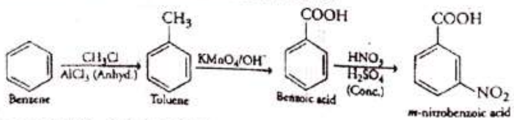
METHLMERCAPTAN (METHYL THIOALCOHOL)

$$CH_3Br + KSH \longrightarrow CH_3SH + KBr$$

6. (b) Prepare the following compounds:

*m-nitro benzoic acid from Benzene * Ethyl alcohol from Molasses

*M-NITRO BENZOIC ACID FROM BENZENE



* ETHYL ALCOHOL FROM MOLASSES

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow[\text{in yeast}]{\text{Sucrase}} C_6H_{12}O_6 + C_6H_{12}O_6$$

Glucose Fructose

$$C_6H_{12}O_6 \xrightarrow[\text{in yeast}]{\text{Zymase}} 2C_2H_5OH + 2CO_2\uparrow$$

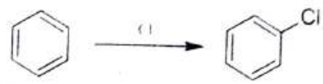
Ethanol

6. (c) Distinguish by simple chemical test

*Hexane and Benzene *Aldehyde and Ketone

HEXANE AND BENZENE

Benzene gives addition reaction with Cl_2 at 50 °C and 400 atm produce Hexachloro cyclo hexane but Hexane is a saturated hydrocarbon, it does not give addition reaction Cl_2.



ALDEHYDE AND KETONE

(See Question No. 2(xiv), 2018)

CHEMISTRY 2012

SECTION 'B'

INORGANIC CHEMISTRY

2. (ii) (a) Write the Valence shell configuration of the following groups.

*IA and IB *VA and VB

Answer:

VALENCE SHELL CONFIGURATIONS

Group I-A: Electronic configuration of valence shell is ns^1

Group I-B: Electronic configuration of valence shell is ,$ns^1,(n-1)d^{10}$.

Group V-A: Electronic configuration of outer valence is ns^2 or np^3

Group V-B: Electronic configuration of valence shell is $ns^2,(n-1)d^3$

2. (iii) Give the reason of the following:

* Ionic Hydrides are called true Hydrides.

Answer: Alkali and Alkaline earth metal hydrides are called true hydrides b/e they are non-volatile & they have high m.p and B.P. They are stable and hence called true hydrides.

* Lithium and Beryllium markedly differ from other members of their respective groups.

Answer: Lithium belongs to group I-A and Beryllium belongs to group II-A of the periodic table their smaller atomic .size results in high charge densities of their ions due to which heat of hydration and electrode potential is high. The Lithium and beryllium markedly differ with other member of their respective group.

* Plaster of Paris is used in making plaster coats and Moulds.

Answer: When $CaSO_4\frac{1}{2}H_2O$ is mixed with water sets in about 5 minutes to a hard mass. This setting takes place with expansion. This property permits to use in the preparation of moulds used in surgery and castings.

* Why is electronic configuration of Chromium (Cr) $4s^1,3d^5$ instead of $4s^2,3d^4$ while that of Copper(Cu) is $4s^1,3d^{10}$ instead of $4s^2,3d^9$?

Answer: The orbitals (sub-shells) tend to become half-filled on completely filled, such configuration are stable than other orbitals. This tendency is called half filled on completely filled orbital rule. Due to this reason electronic configuration of Chromium (Cr) is $4s^1, 3d^5$ instead of $4s^2, 3d^4$. Similarly electronic configuration of Copper(Cu) is $4s^1, 3d^{10}$ instead of $4s^2, 3d^9$

2. (iv) Give equations for the following

* Zinc is treated with 90% Conc H_2SO_4
* Action of Heat on $KMnO_4$
* Aluminum reacts with aqueous Sodium Hydroxide
* Lunar Caustic is heated 450°C

ANSWER

(i) $4Zn + 5H_2SO_4 \xrightarrow{90\%} 4ZnSO_4 + H_2S + 4H_2O$

(ii) $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$

(iii) $2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$

(iv) $2AgNO_3 \xrightarrow{450°} Ag + 2NO_2 + O_2$

2. (v) (b) Write I.U.P.A.C. names of the following

*$[Fe(NO_2)_6]^{3-}$ *$[Cr(en)_2Cl_2]Cl$

ANSWER

*$[Fe(NO_2)_6]^{3-}$

Hexa nitro ferrate (III) ion

*$[Cr(en)_2Cl_2]Cl$

Bis (ethylene diamine) dichlorochromium (III) chloride

2. (viii) Complete the following equation and give I.U.P.A.C. names of the complexes formed

*$CuSO_4 + NH_3 \xrightarrow{excess}$ *$AgNO_3 + NH_3 \xrightarrow{excess}$

I.U.P.A.C. names of the complexes is Tetra Amino Cuprous (II) Sulphate

$$AgNO_3 + 2NH_3 \xrightarrow{excess} [Ag(NH_3)_2]NO_3$$

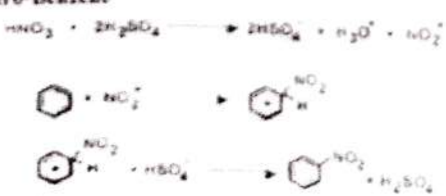
I.U.P.A.C. names of the complexes is Diamine Silver Nitrate

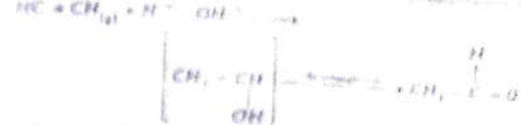
ORGANIC CHEMISTRY

2. (x) Write equation only for the for the following conversions

*Benzene to Nitro-Benzene *Ethyie to Ethanal *Ethanol to Ethanoic acid

*Benzene to Nitro-Benzene





*Ethanol to Ethanoic acid

$$CH_3CH_2OH \xrightarrow{K_2Cr_2O_7/H_2SO_4} CH_3CHO$$

$$CH_3CHO \xrightarrow{K_2Cr_2O_7/H_2SO_4} CH_3COOH$$

2. (xiii) (b) Give chemical test to distinguish between the following

*Ethene and Ethyne

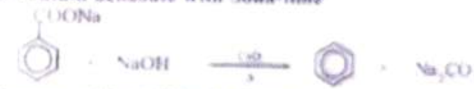
(See Question No. 2(xiv), 2018)

*Aldehyde & Ketone

(See Question No. 2(xiv), 2018)

2. (xv) Give equations for the following reactions:

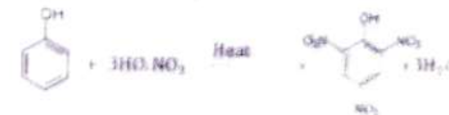
* Reaction of Sodium benzoate with Soda-lime



Reaction of Acetone with acidified $K_2Cr_2O_7$

Reaction of Sodamide with Chloroethane

* Reaction of Phenol with Conc. HNO_3



SECTION 'C'

INORGANIC CHEMISTRY

4. (b) Complete the following equations.

Answer:

(i) $4Zn + HNO_3 \xrightarrow[Dilute]{Cold} 4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$

(ii) $Ca_2B_6O_{11} + 2Na_2CO_3 \rightarrow 2CaCO_3 + Na_2B_4O_7 + 2NaBO_2$

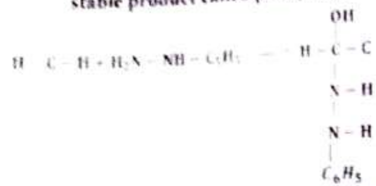
(iii) $Cl_2 + 2NaOH \xrightarrow[Dilute]{Cold} NaCl + NaClO + H_2O$

(iv) $2CaSO_4 \cdot 2H_2O \xrightarrow{100°C} (CaSO_4)_2 \cdot H_2O + 3H_2O$

ORGANIC CHEMISTRY

5. (b) What happens when:

* Methanal is treated with Phenyl hydrazine.
When Methanal is treated with Phenyl hydrazine in normal manner to form stable product called phenyl hydrazine

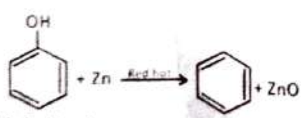


Unstable Addition Prodct

Phenyl Hydrazone

* **Phenol is treated with Bromine water.**
When Phenol reacts rapidly with bromine water and produces 2,4,6-tri bromo phenol

* **Phenol is treated with red hot Zinc dust.**
When vapours of phenol are passed over red hot zinc dust then it is converted into benzene



* Ethyne is reacted with Iodine in presence of Ethanol
When Ethyne is reacted with Iodine in presence of Ethanol it yields Acetylene di-Iodide

(i) $CH_2 = \underset{\underset{CH_3}{|}}{C} - C \equiv CH$

(ii) $CH_3 - \overset{\overset{O}{\|}}{C} - O - \overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}} - CH_3$

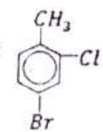
(iii) $CH_3 - CH = CH - CH_2 - COOH$

(iv) $CH_3 - CH = CH - \underset{\underset{OH}{|}}{CH} - \underset{\underset{CH}{|}}{CH} - CH_3$

(v) $CH_3 - CO - CH_2 - \underset{\underset{CH_3}{|}}{CH} - CH_3$

(vi) $CH_3 - CH_2 - O - \underset{\underset{CH_3}{|}}{CH}$

(vii)



ANSWER:

(i) 2-methyl-1-butene-3-yne
(ii) 2- methyl-4-hexen-3-ol
(iii) Terbutyl ethanoate
(iv) 4-methyl-2-pentanoate
(v) 3-pentenoic acid
(vi) 2-ethoxy propane
(vii) 4-bromo-2-chlorotoluene

CHEMISTRY 2011

SECTION 'B'

INORGANIC CHEMISTRY

2. (i)(a) Write the electronic configuration, group, period and block of elements with atomic number 24 and 29.

Answer

Atomic No.	electronic configuration	Group	Period	Block
24	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^5$	V-I B	"4"	"d"
29	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^{10}$	V-III B	"4"	"d"

2. (v)Give reasons:

* **Alkaline earth metal ions are more strongly hydrated than alkali metal ions.**

Answer: Alkaline earth metals are more strongly hydrated than alkali metals ions because M^{+2}ions have a strong electric field, more nuclear charge and small size as a result II A metal ions are hydrated strongly than IA ions.

Atomic Hydrogen is more reactive than mole...

Answer: Atomic Hydrogen has one electron in its outermost orbit. Thus it can accept or donate one electron to reach the electronic configuration of the nearest inert gas. On the other hand, molecular hydrogen consists of two atoms of hydrogen linked together by a covalent bond. Thus, there is no extra electron for reactivity. For the above reason, atomic hydrogen can readily react with little amount of energy. Molecular hydrogen requires greater amount of energy so as to form atoms of hydrogen first which will react.

* Most of the transition elements and their compound are paramagnetic.

Answer: The transition metals and their compounds show magnetic properties. Many compounds of the transition metals are paramagnetic because of unpaired electron spins in the atom.

* H_2O and NH_3, act as Ligands but H_3O^+ & NH_4^+ do not

Answer: In H_2O and NH_3, there is a lone pair of electron which is available for donation. Therefore they act as ligands while in a case of H_2O^+ and NH_4^+ Ion pair of electron is not available-in fact it is consumed in making of coordinate covalent bond.

2. (vi)Give equations for the following:

* Reaction of Na_2CO_3 with Silica

$$Na_2CO_3 + SiO_2 \rightarrow 7Na_2SiO_3 + CO_2$$

* Reaction between Sodium and Oxygen

$$2Na + O_2 \rightarrow Na_2O_2$$

* Action of heat on Blue vitriol

$$CuSO_4.5H_2O \rightarrow CuSO_4 + 5H_2O$$

* Reaction between K_2CrO_7, and KOH

$$K_2Cr_2O7 + 2KOH \rightarrow 7\ 2K_2CrO_4 + H_2O$$

2. (vii) (a) Give the equations for the preparation of the following;

*Bleaching Powder *Boric acid

BLEACHING POWDER

$$Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$$

BORIC ACID

$$BN + {}_3H_2O \rightarrow H3BO_3 + NH_3$$

2. (vii) (b) Give points of resemblance of Hydrogen with IA and VII-A groups of periodic table.

Answer: (See Question No. 3 (c) 2018)

2. (viii) Name the complexes by I.U.P.A.C. system:

*$[Ni(CO)_4]$ *$NH_4[Cr(NCS)_4(NH_3)_2]$ *$[Cr(NH_3)_6]$

*$[Fe(CN)_6]^{-3}$

ANSWER:

*$[Ni(CO)_4]$

Tetra Carbonyl Nickel (0)

*$NH_4[Cr(NCS)_4(NH_3)_2]$

Ammonium diamminetetrathiocyanato-S Chromate (III)

*$[Cr(NH_3)_6]$

Hexa ammine chromium (III) ion

*$[Fe(CN)_6]^{-3}$

Hexa Cyano Ferrate (III) ion

2. (ix) (b) Distinguish by simple chemical tests:

*Aldehyde and Ketone

(See Question No. 2(xiv), 2018)

*Alkene and Alkyne

(See Question No. 2(xiv), 2018)

2. (x) Define Isomerism. Give its types with examples.

(See Question No. 2 (ix) 2018)

2. (xi) Give the equations for the preparation of the following:

* Acetaldehyde form Acetylene

$$H - C \equiv C - H + H_2O \xrightarrow[HgSO_4]{20\%\ H_2SO_4} > CH_3CHO$$

* Ethene from Ethyl alcohol

$$C_2H_5OH + H_2SO_4 \rightarrow C_2H_5.O.SO_2.OH + H_2O$$

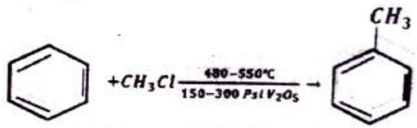
$$C_2H_5.O.SO_2.OH \xrightarrow{Heat} > C_2H_{4(g)} + H_2SO_{4(aq)}$$

* Oxalic acid from Ethyne

$$HC \equiv CH + 2[O] \xrightarrow{KMnO_4} > OH - \overset{\overset{O}{||}}{C} - \overset{\overset{O}{||}}{C} - OH$$

*Benzene from Toluene

When Benzene reacts with Methyl chloride in the presence of V_2O_5 at 480-550°C and 150-300 Psi it yield Toluene



SECTION 'C'

ORGANIC CHEMISTRY

5. (a) Give equations for the preparation of following organic compounds: (i)Ethyne from Ethene (ii)Diethyl ether from Ethanol (iii)Methanol from Water gas (iv)Methanal from Methanol (v)Acetone from Acetic acid (vi)Ethanoic acid from Ethanol (vii) Ethanol from Glucose

(i)Ethyne from Ethene

$$H_2 - C = C - H_2 \longrightarrow > H - C \equiv C - H$$

(ii)Diethyl ether from Ethanol

$$CH_3 - CH_2^+ - OH^- + O^-CH_2 - CH_3 \xrightarrow{H_2SO_4} > CH_3 - CH_2 - O - CH_2 - CH_3$$

(iii)Methanol from Water gas

$$\left[CO_{(g)} + H_{2(g)} \xrightarrow[400-450°C/200atm]{ZnO/Cr_2O_3} > CH_3OH\right]$$

(iv)Methanal from Methanol

$$O = C - H_2 + 2[H] \xrightarrow[C_2H_5OH]{Na} > CH_3OH$$

(v)Acetone from Acetic acid

$$CH_3 - \overset{O}{\overset{||}{C}} - OH \xrightarrow[500^\circ C]{MnO_2} CH_3 - \overset{O}{\overset{||}{C}} - CH_3 + CO_2 + H_2O$$

(vi)Ethanoic acid from Ethanol

$$CH_3CH_2OH \xrightarrow[[O]]{K_2Cr_2O_7/H_2SO_4} > CH_3CHO$$
$$CH_3CHO \xrightarrow[[O]]{K_2Cr_2O_7/H_2SO_4} > CH_3COOH$$

(vii) Ethanol from Glucose

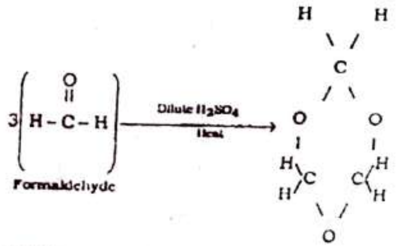
$$C_6H_1O_6 \xrightarrow[In\ Yeast]{Zymse} > 2C_2H_5OH + 2CO_2 \uparrow$$

5. (b) Give equation for the following reaction:
(i)Cannizzaro's Reaction (ii)Polymerization of Acetaldehyde (iii) Reduction of Acetone (iv)Esterification (v)Combustion of Ethane (vi)Oxidation of Benzene(vii) Hydrolysis of Ethyl acetate in basic medium

(i)Cannizzaro's Reaction

$$H - CHO + H - CHO + NaOH \longrightarrow > CH_3 - OH + HCO\bar{O}Na^+$$

(ii)Polymerization of Acetaldehyde



(iii) Reduction of Acetone

$$CH_3 - \overset{O}{\overset{||}{C}} - CH_3 + H_2 \xrightarrow[Heat]{Pd} > CH_3 - \overset{O}{\overset{||}{CH}} - CH_3$$

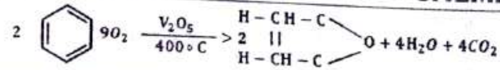
(iv)Esterification

$$CH_3 - \overset{O}{\overset{||}{C}} - \boxed{OH + H}O - C_2H_5 \xrightarrow[Heat]{H_2SO_4} > CH_3 - \overset{O}{\overset{||}{C}} - O - C_2H_5 + H_2O$$

(v)Combustion of Ethane

$$2C_2H_6 + 7O_2 \xrightarrow[75^\circ C]{HgSO_4/H_2SO_4} > 4CO_2 + 6H_2O + Heat$$

(vi)Oxidation of Benzene



(vii) Hydrolysis of Ethyl acetate in basic medium

$$CH_3 - \overset{O}{\overset{||}{C}} - OC_2H_5 + NaOH(10\%) \longleftrightarrow CH_3 + \overset{O}{\overset{||}{C}} - ONa + C_2H_5OH$$

6. (b) Write the I.U.P.A.C. names of the following organic compounds:

(i) $CH_3 - CH_2 - CH = CH - COOH$

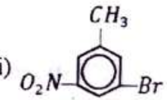
(ii) $CH_2 = CH - C \equiv C - CH - CH_2$

(iii) $CH_3 - \overset{O}{\overset{||}{C}} - O - C_6H_5$

(iv) $CH_3 - O - CH_2 - \overset{OCH_3}{\overset{|}{CH}} - CH_2 - OC_2H_5$

(v) $(CH_3)_2\underset{CH_3}{\underset{|}{CH}} - \overset{O}{\overset{||}{C}} - CH(C_2H_5)_2$

(vi) $H - \overset{O}{\overset{||}{C}} - (CH_2)_2 - \overset{O}{\overset{||}{C}} - H$

(vii) 

ANSWER:

(i) 2-buten-oic acid
(ii) 1,5-hexadien-3-yne
(iii) Phenyl ethanoate
(iv) 1-ethoxy-2,3-dimethoxypropane
(v)4-ethyl-2-methyl-3-hexanone
(vi) Heptandioic acid
(vii) 3-bromo-5-nitro Toluene

THE END

SUFFA ACADEMY